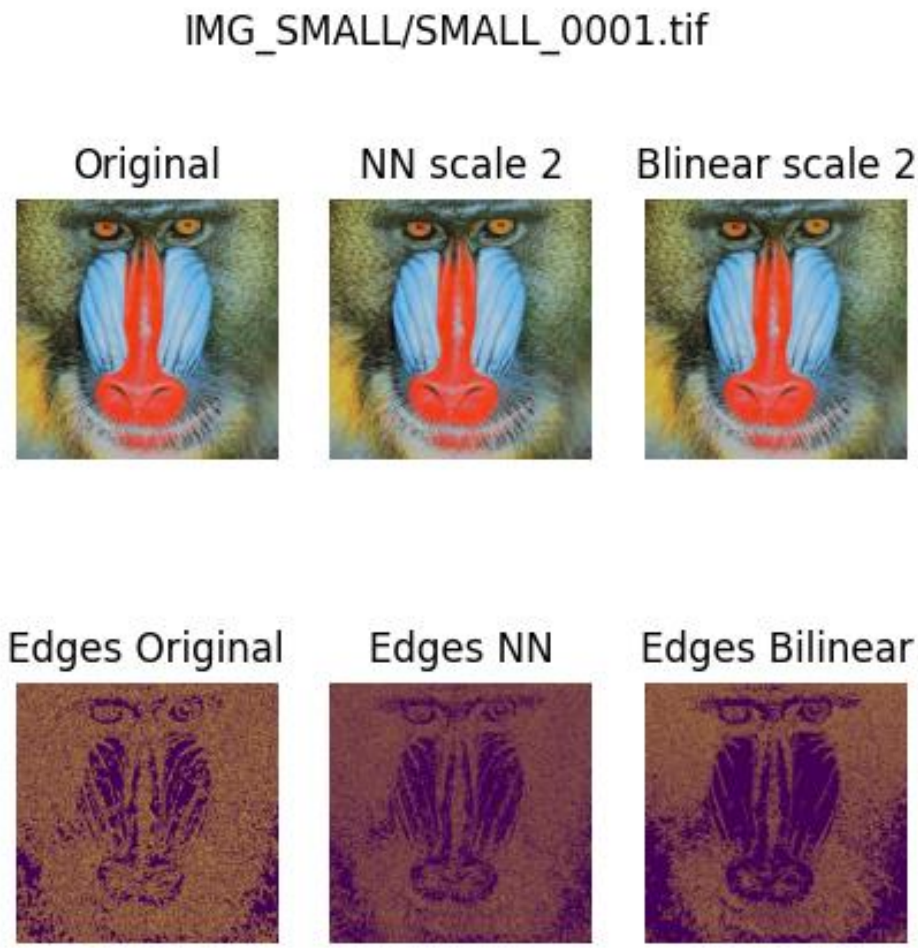
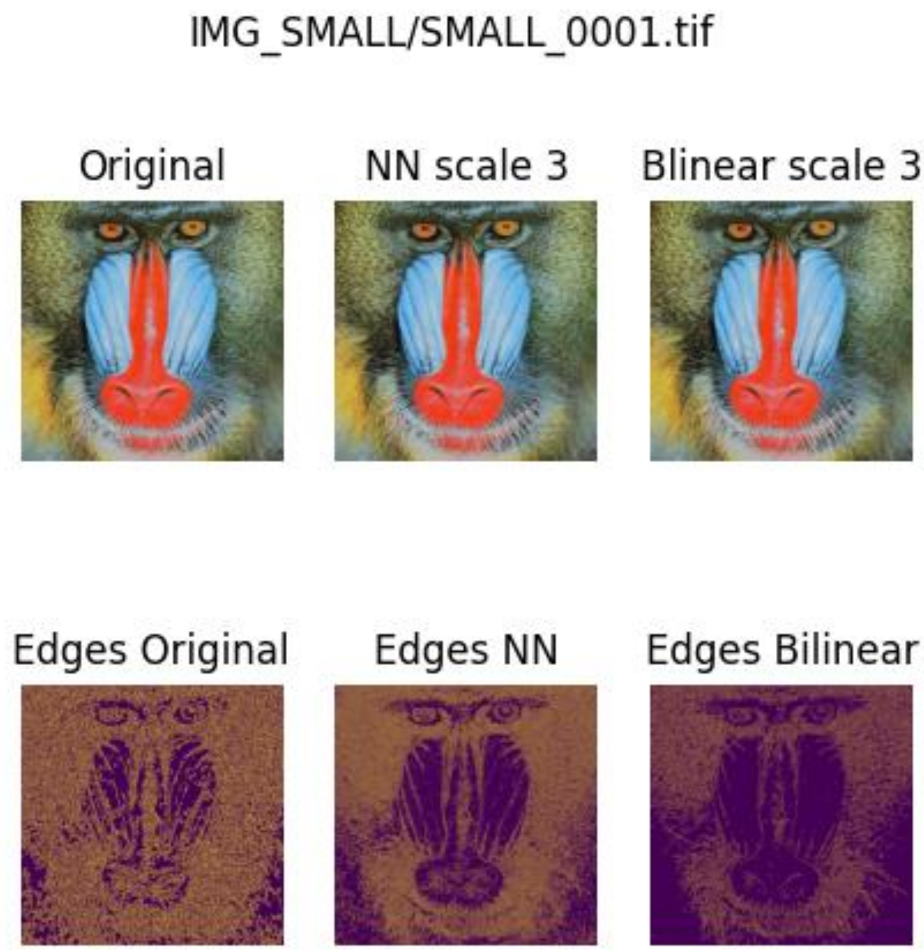
Report

Autor:

Mateusz Hypś

Test algorytmów powiększania

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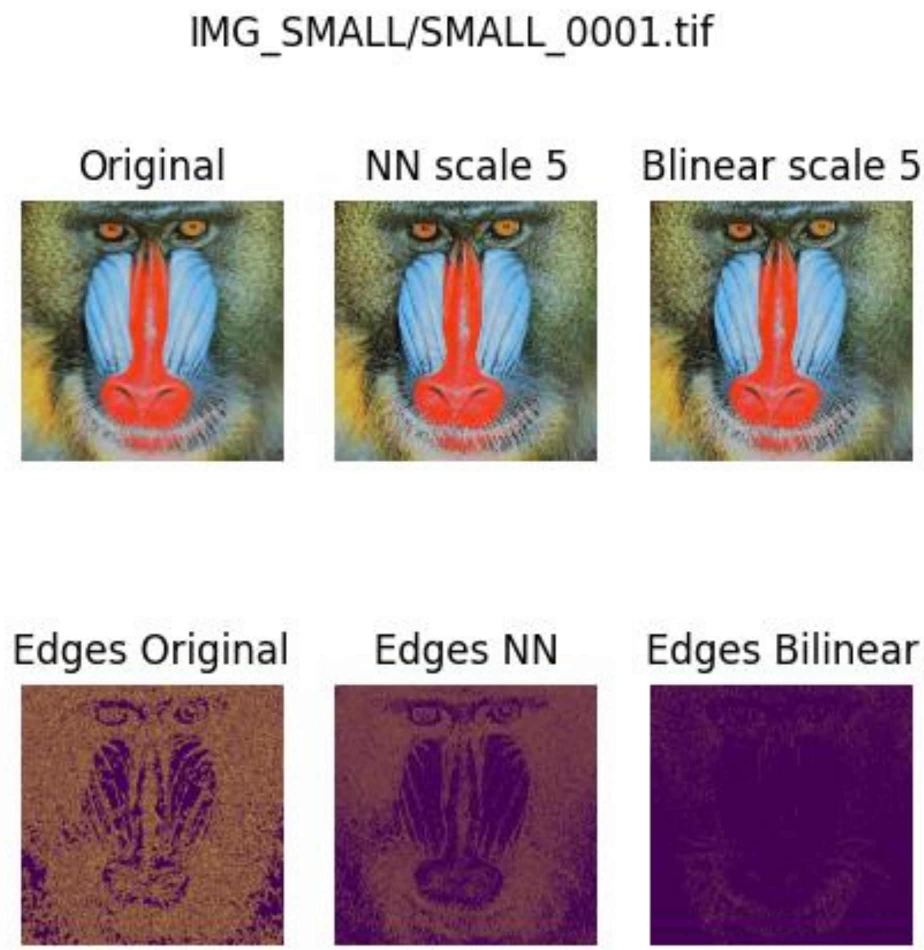
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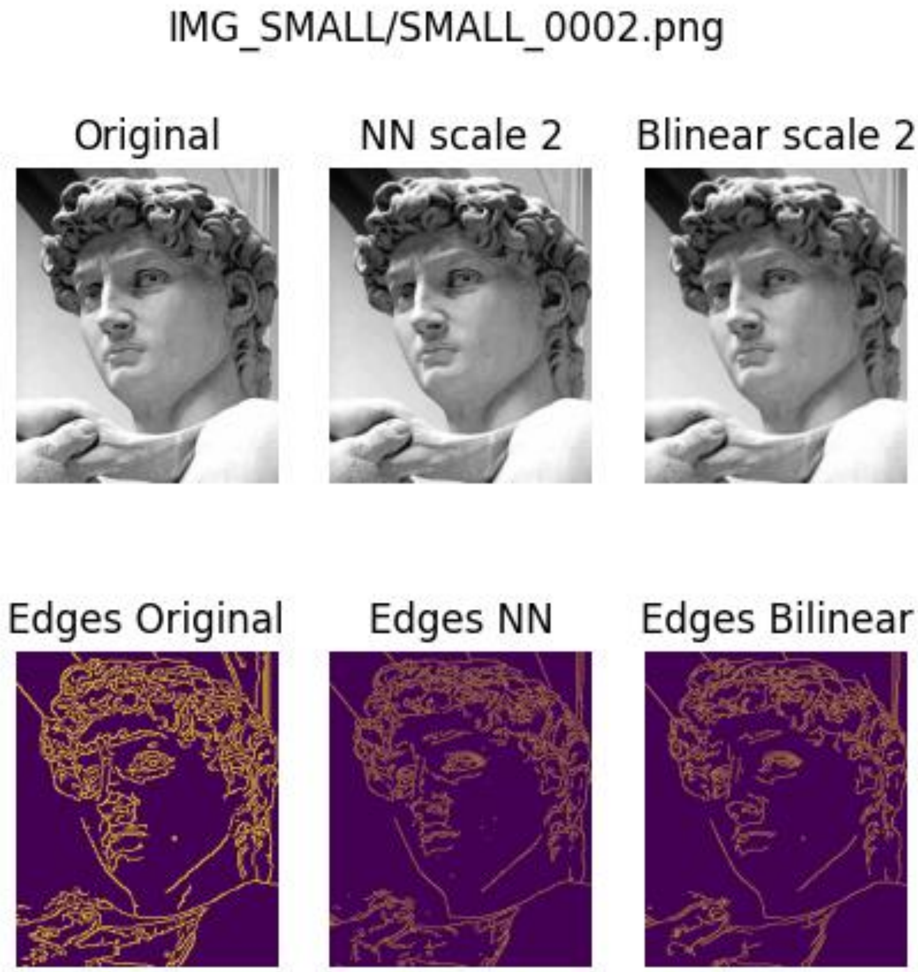
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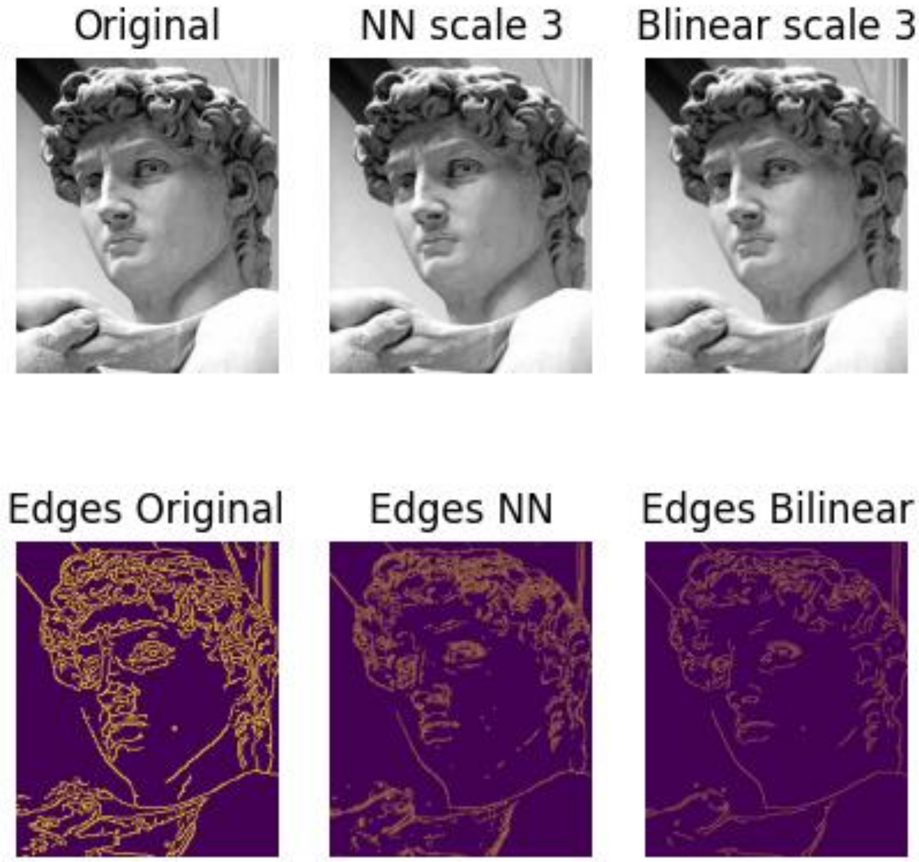

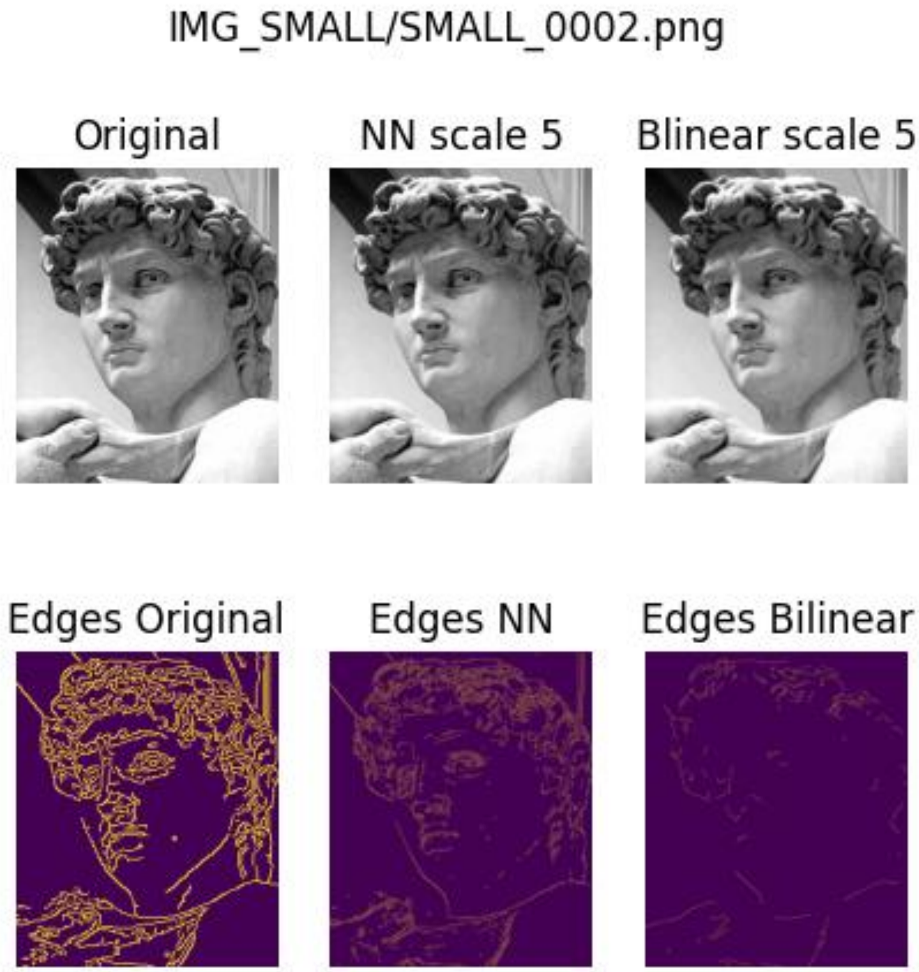
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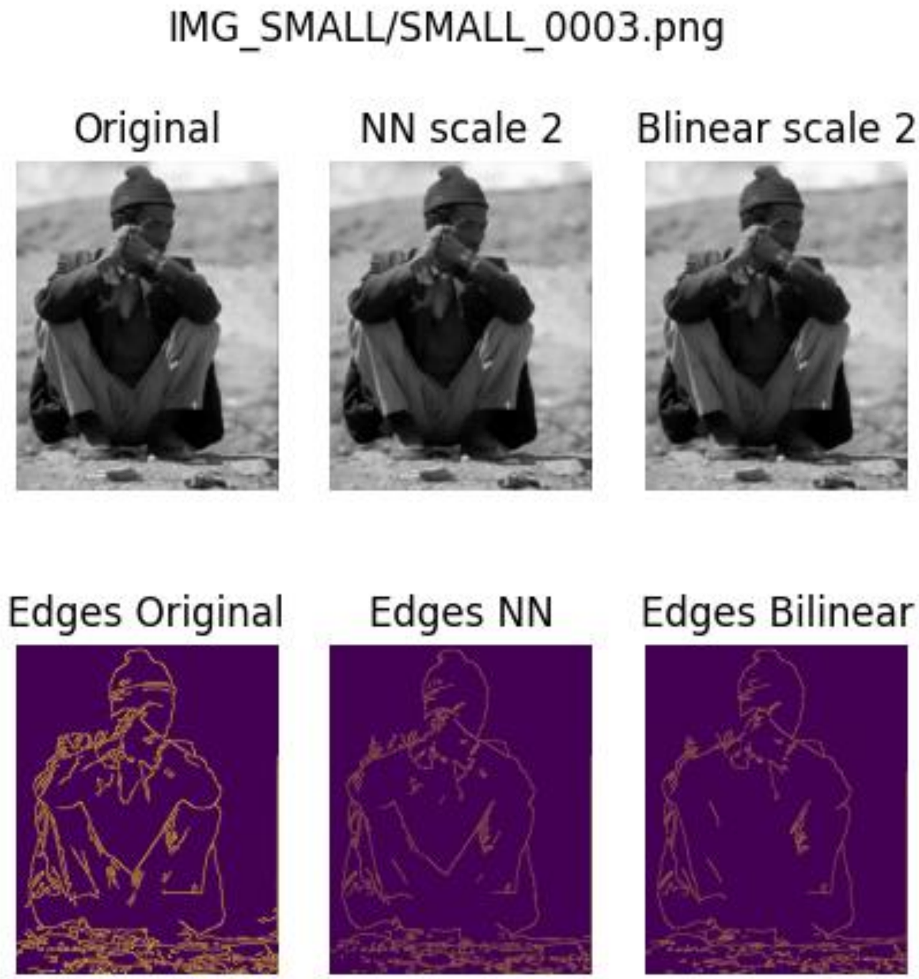
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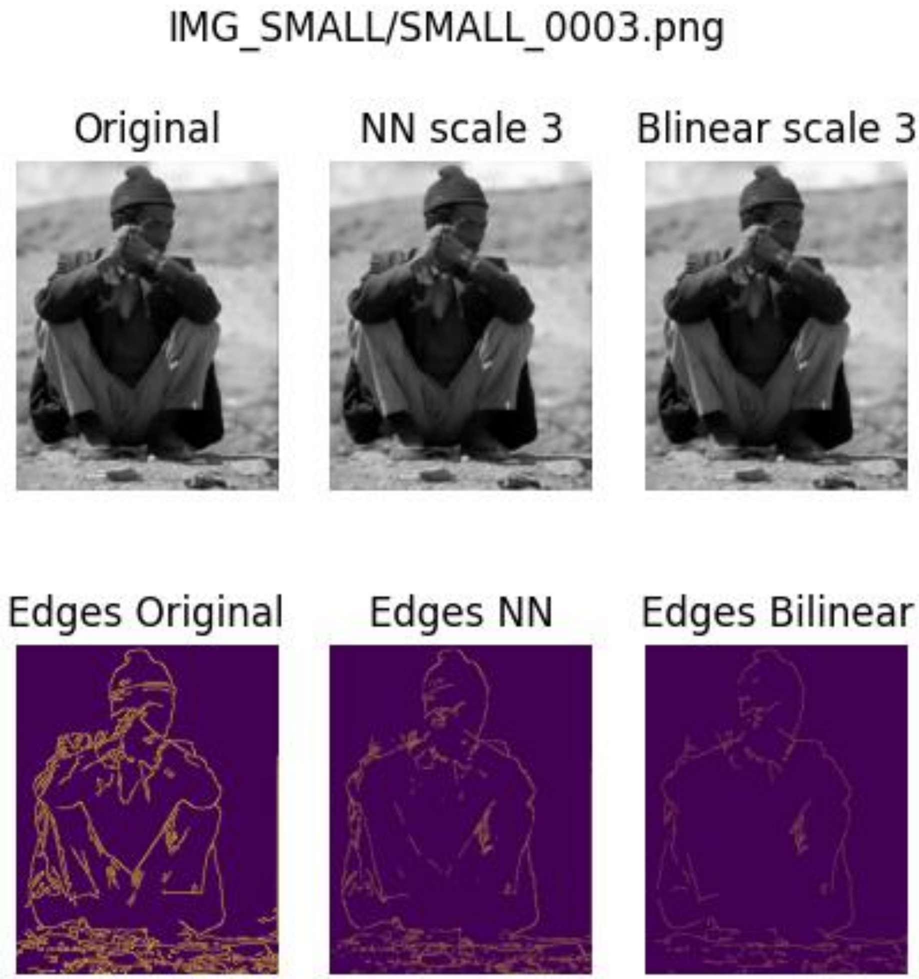
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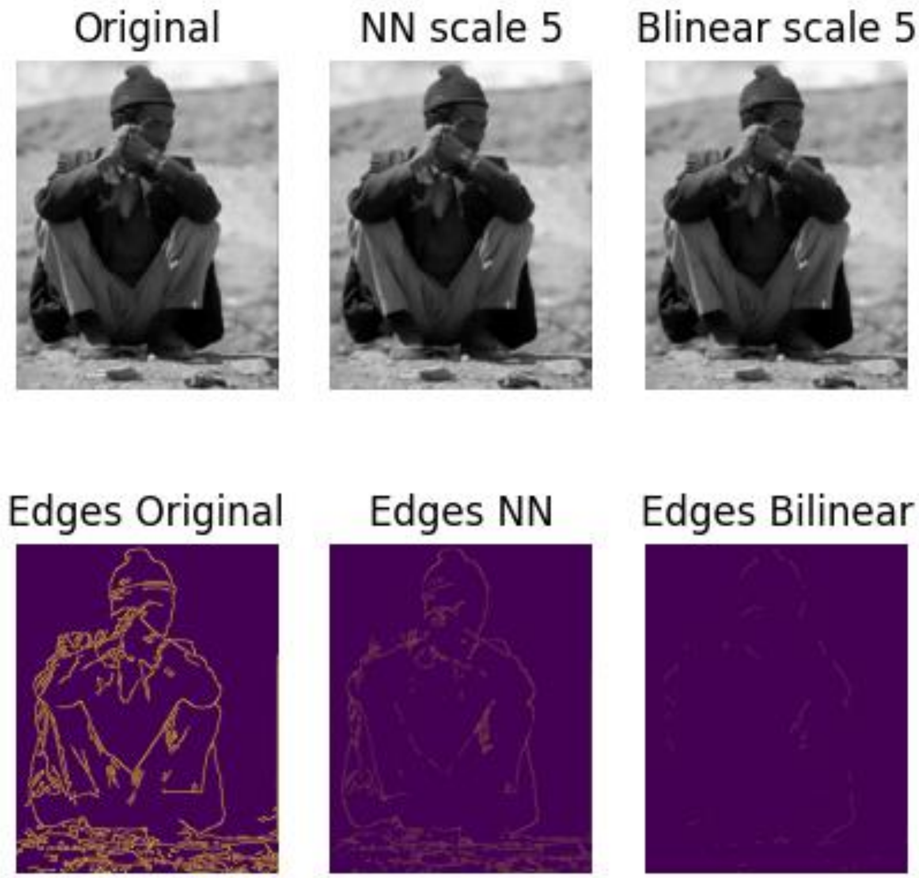

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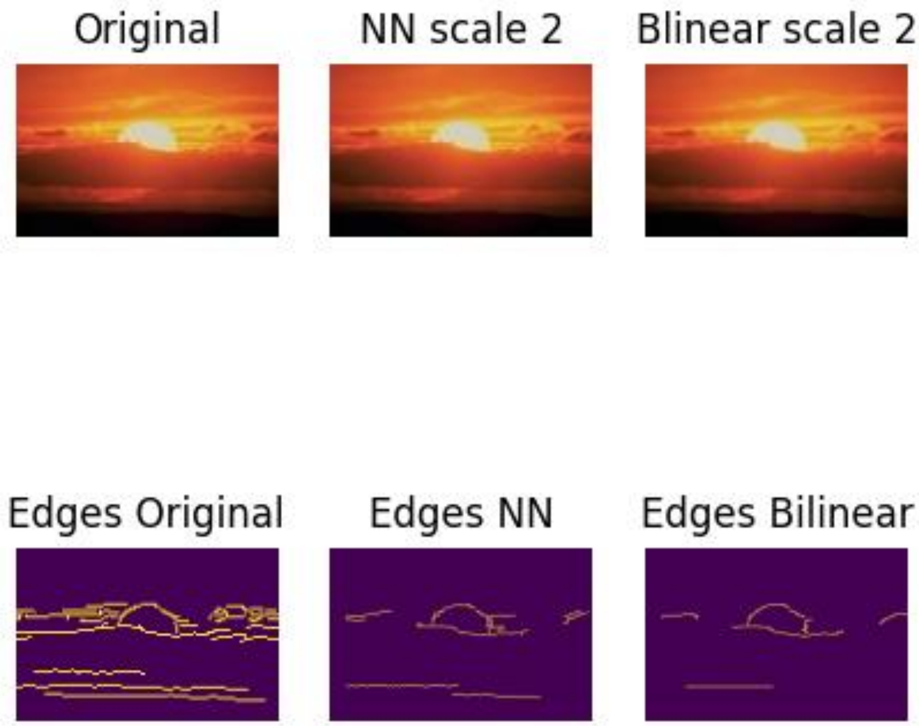

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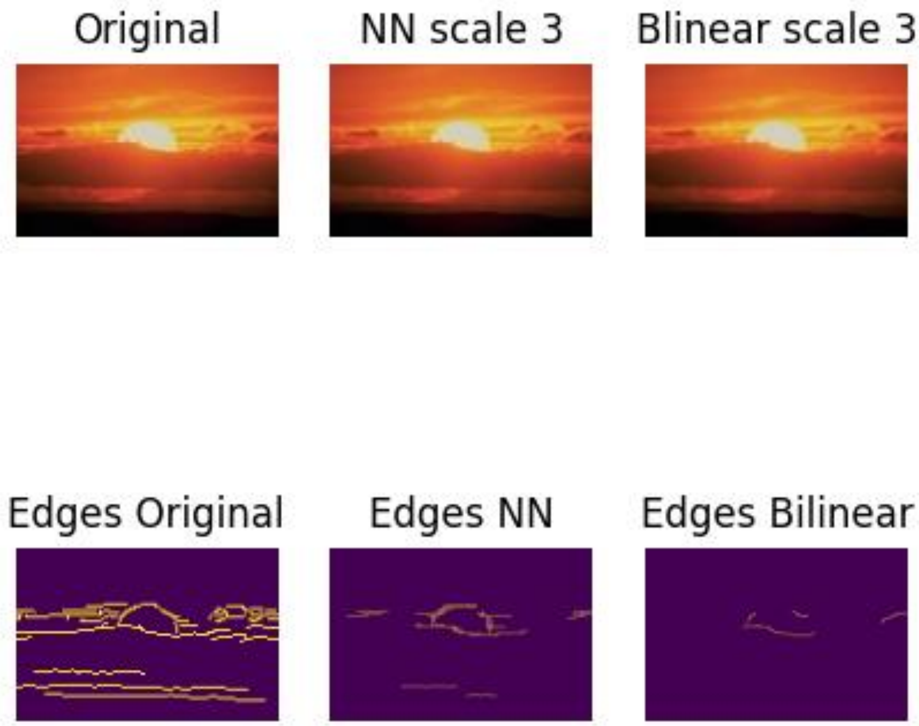

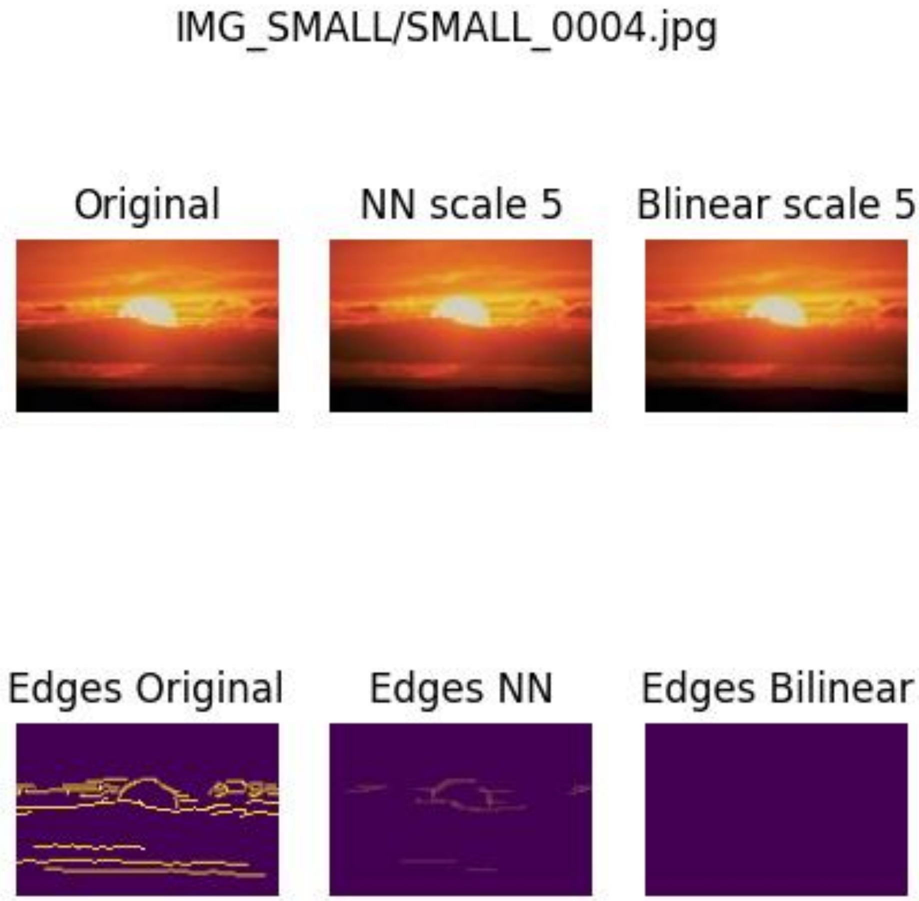
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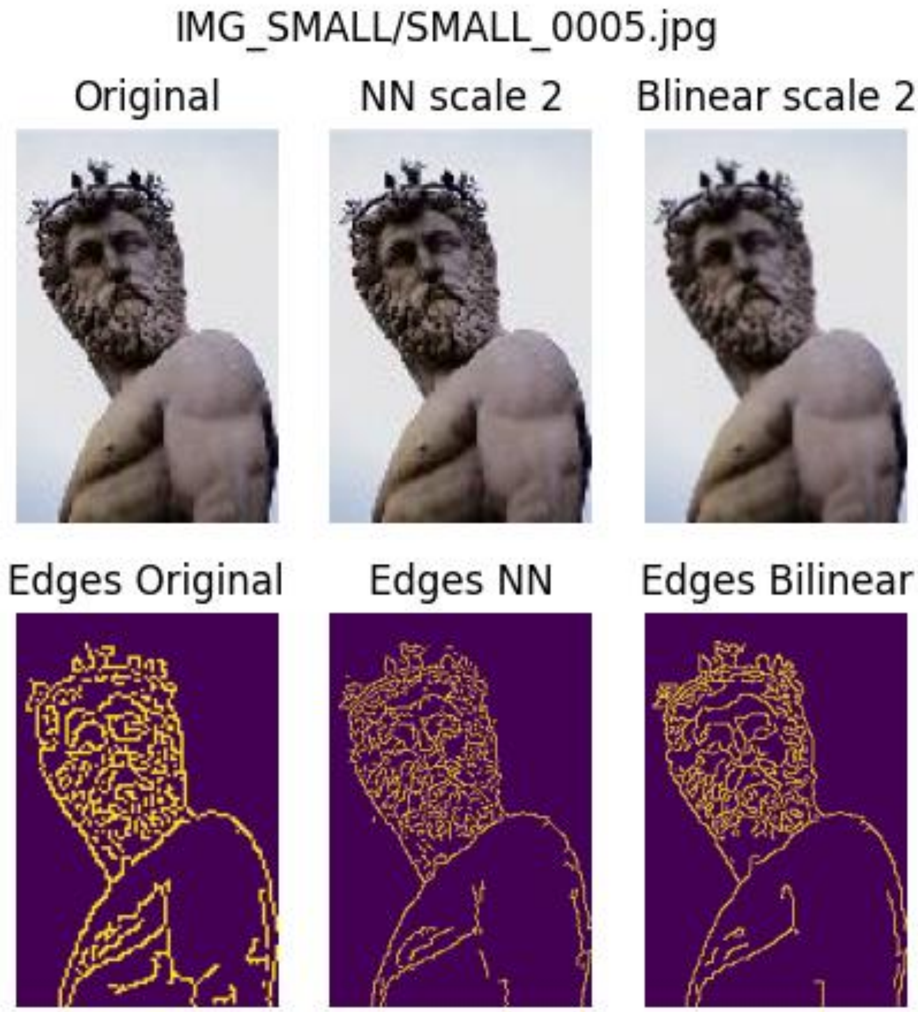
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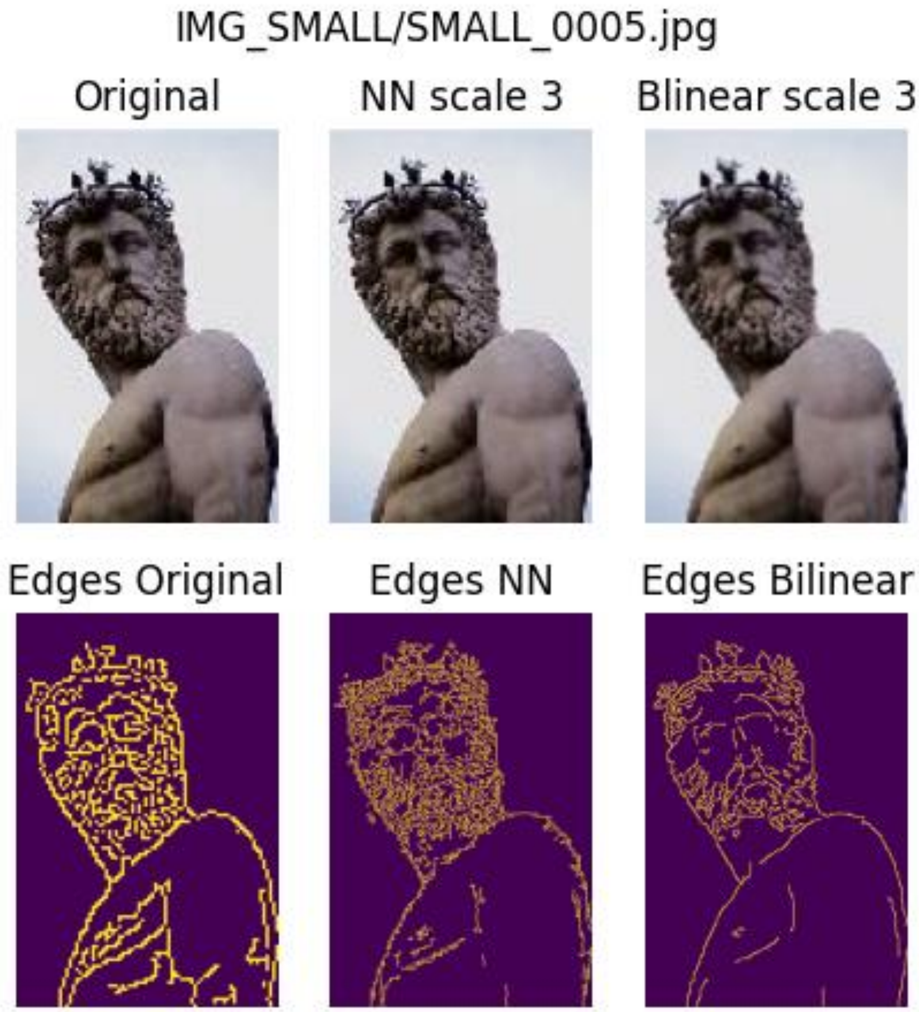
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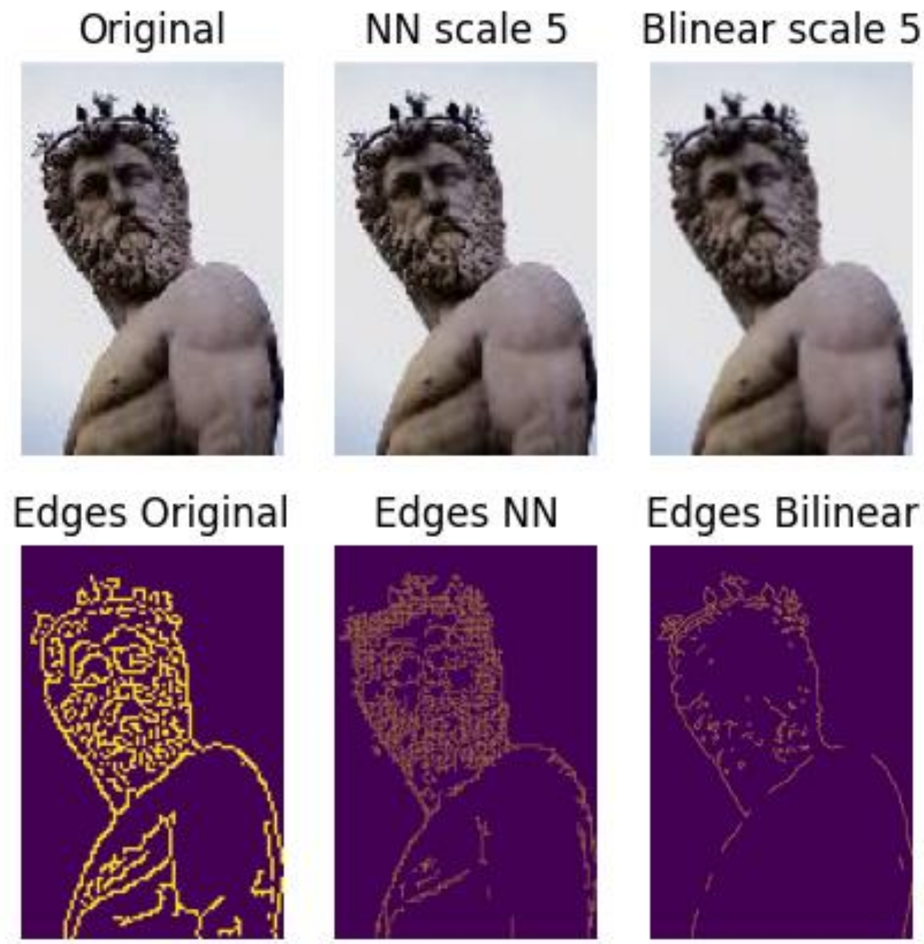

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Edges Original | Edges NN | Edges Bilinear

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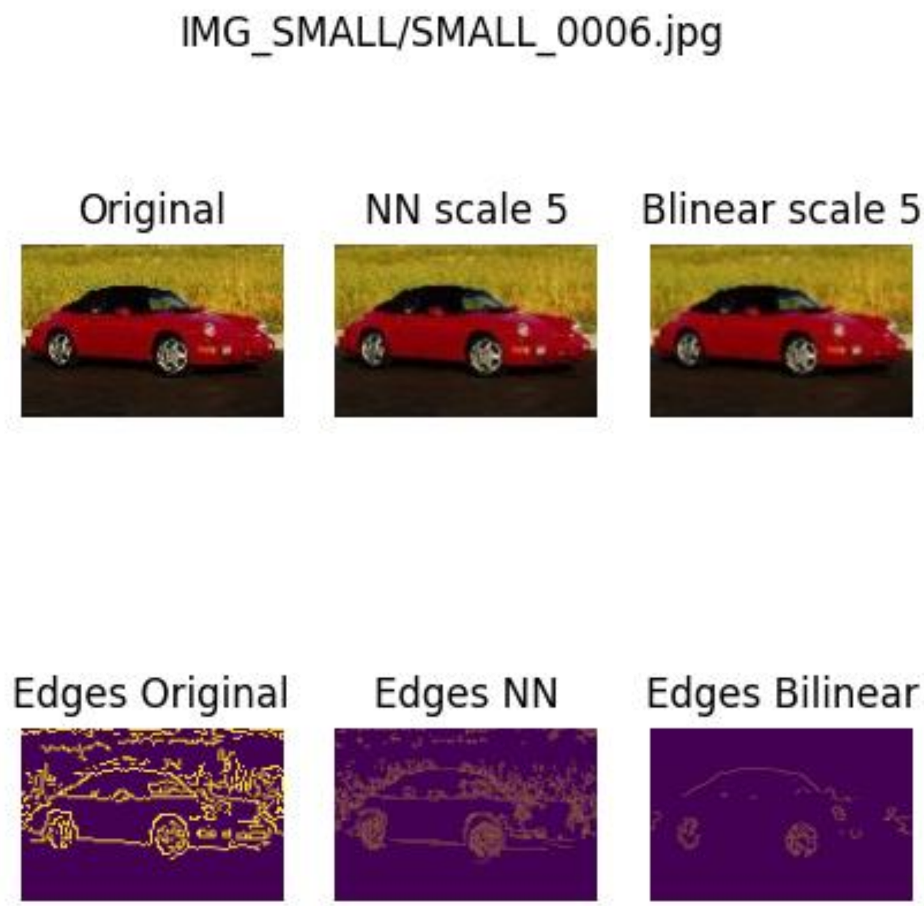
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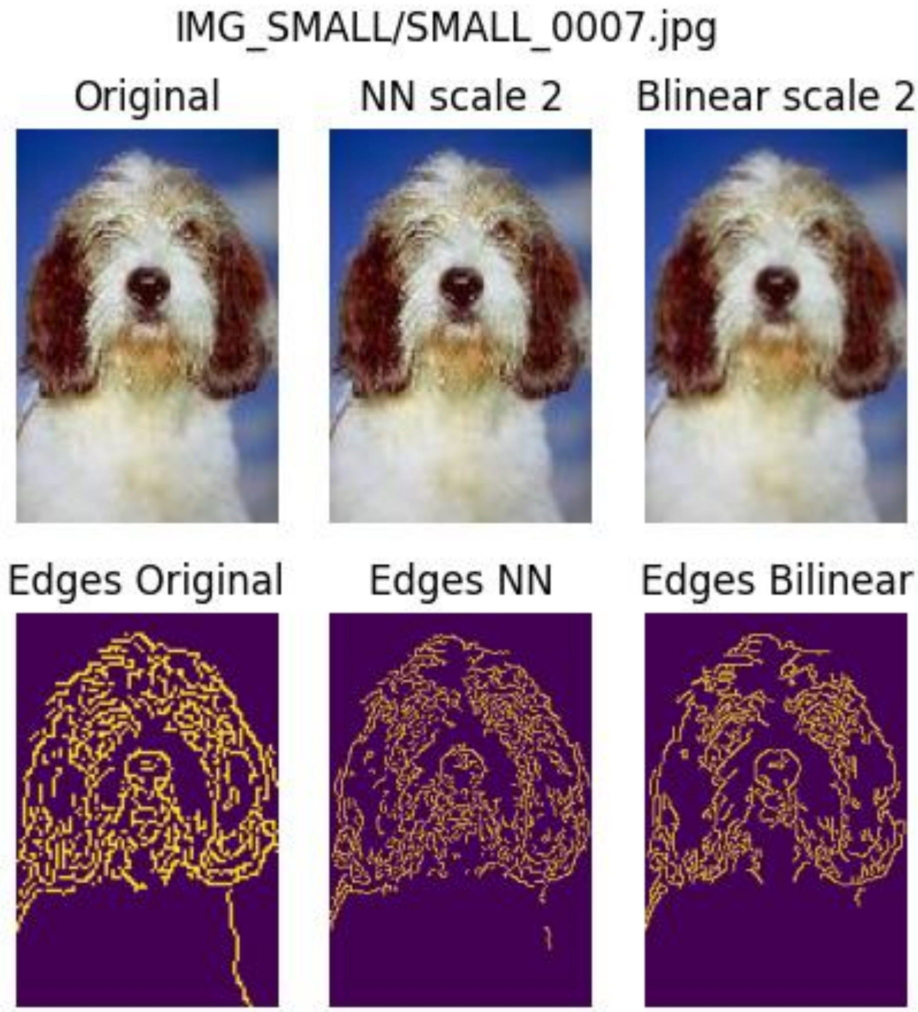
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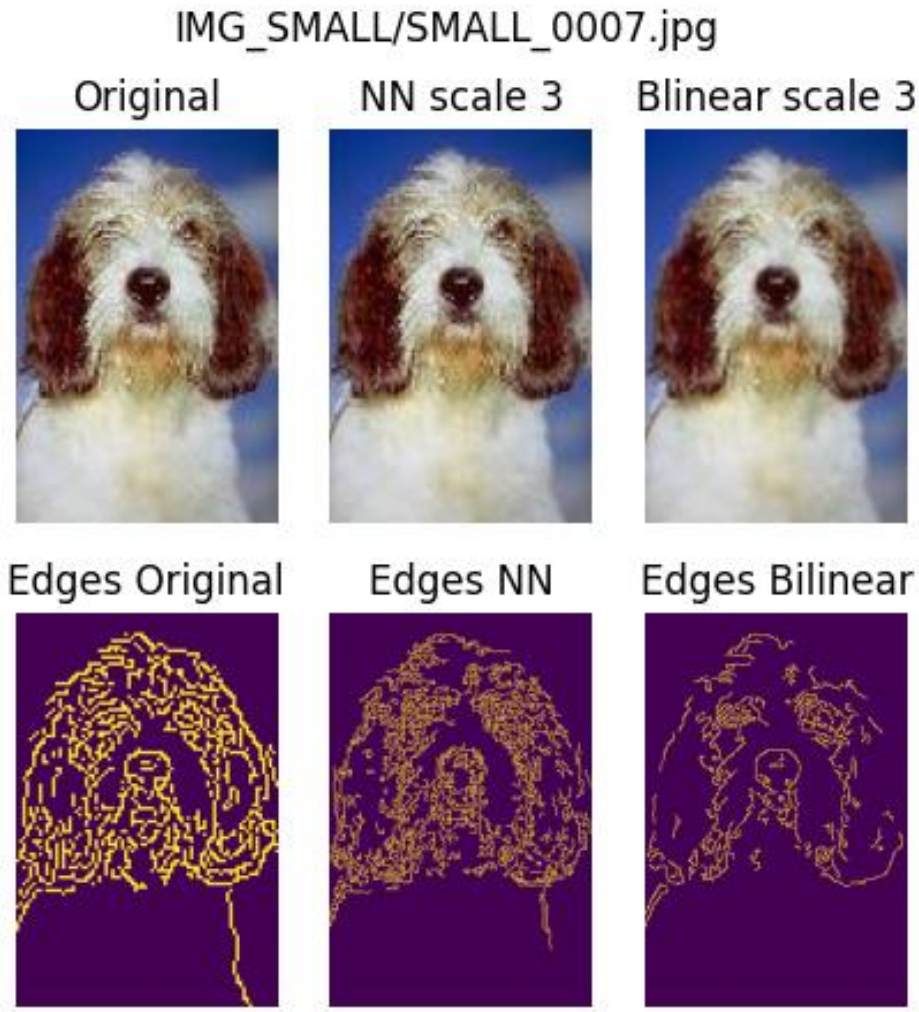
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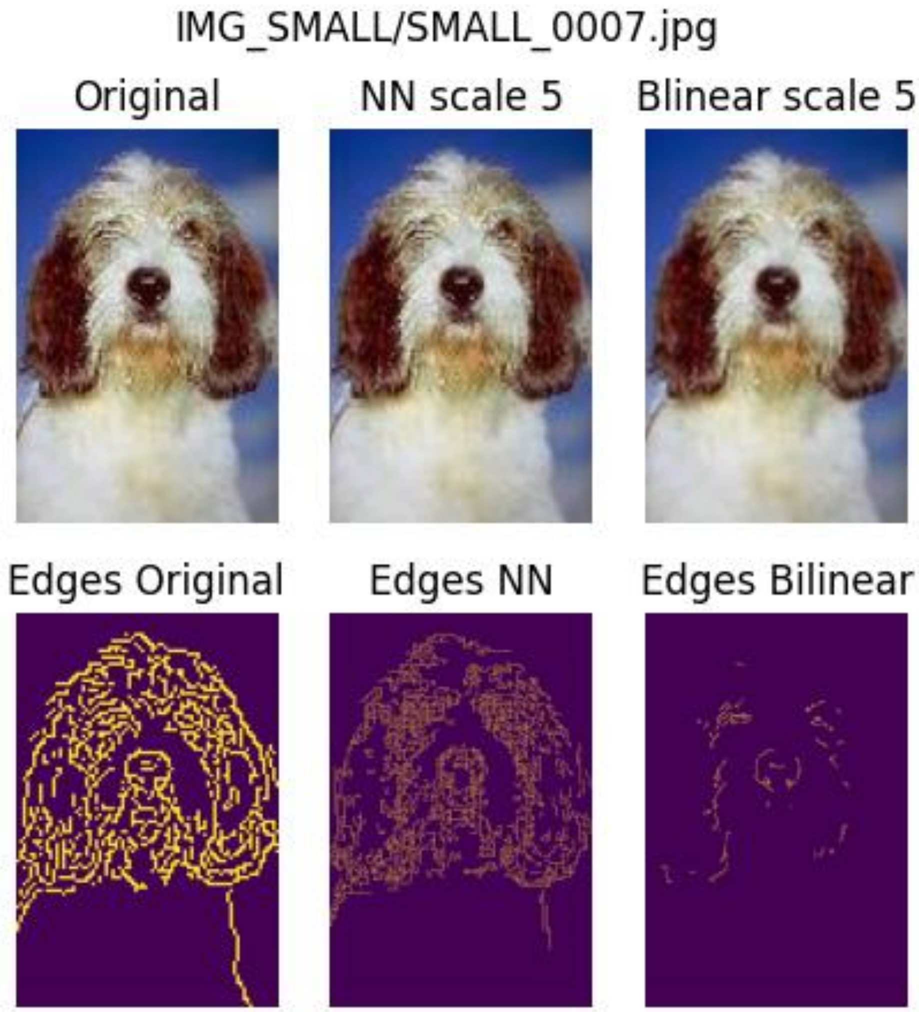
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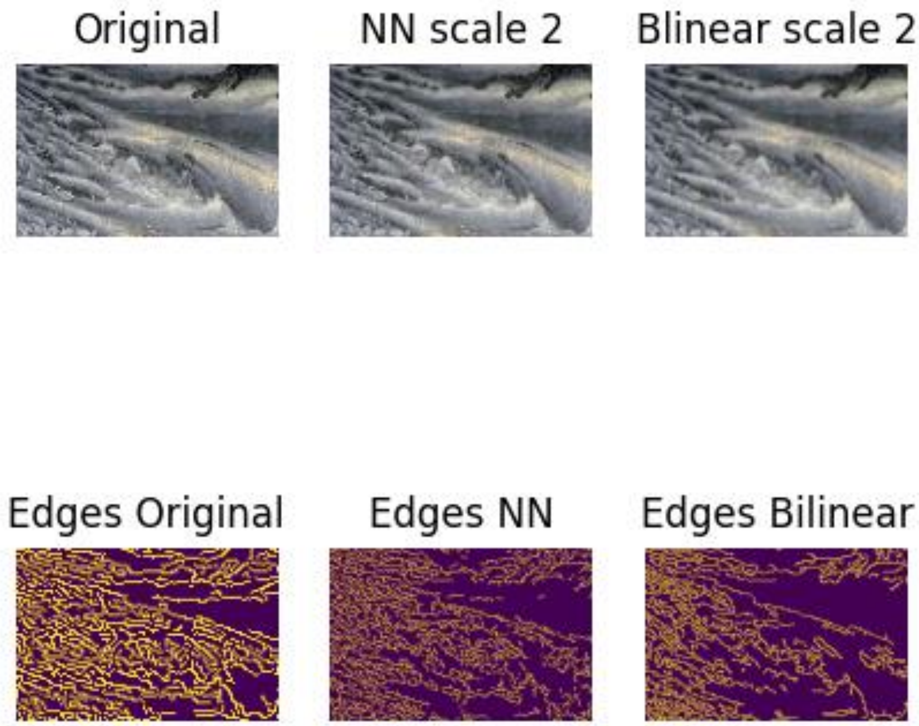

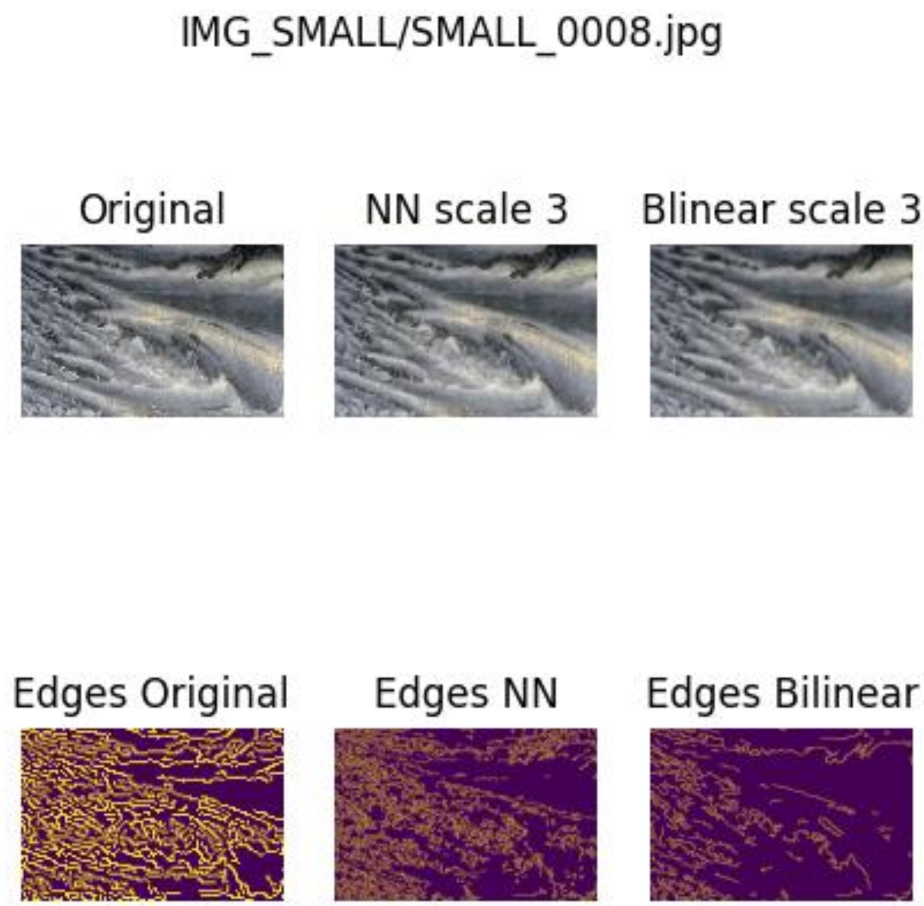
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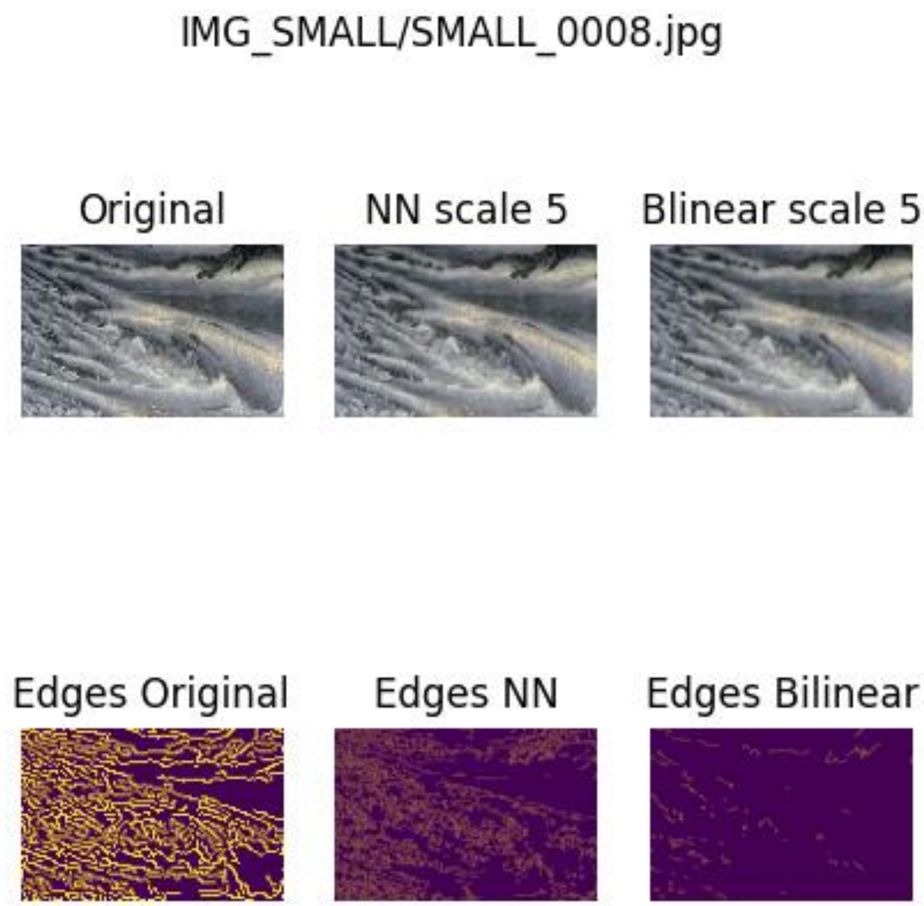
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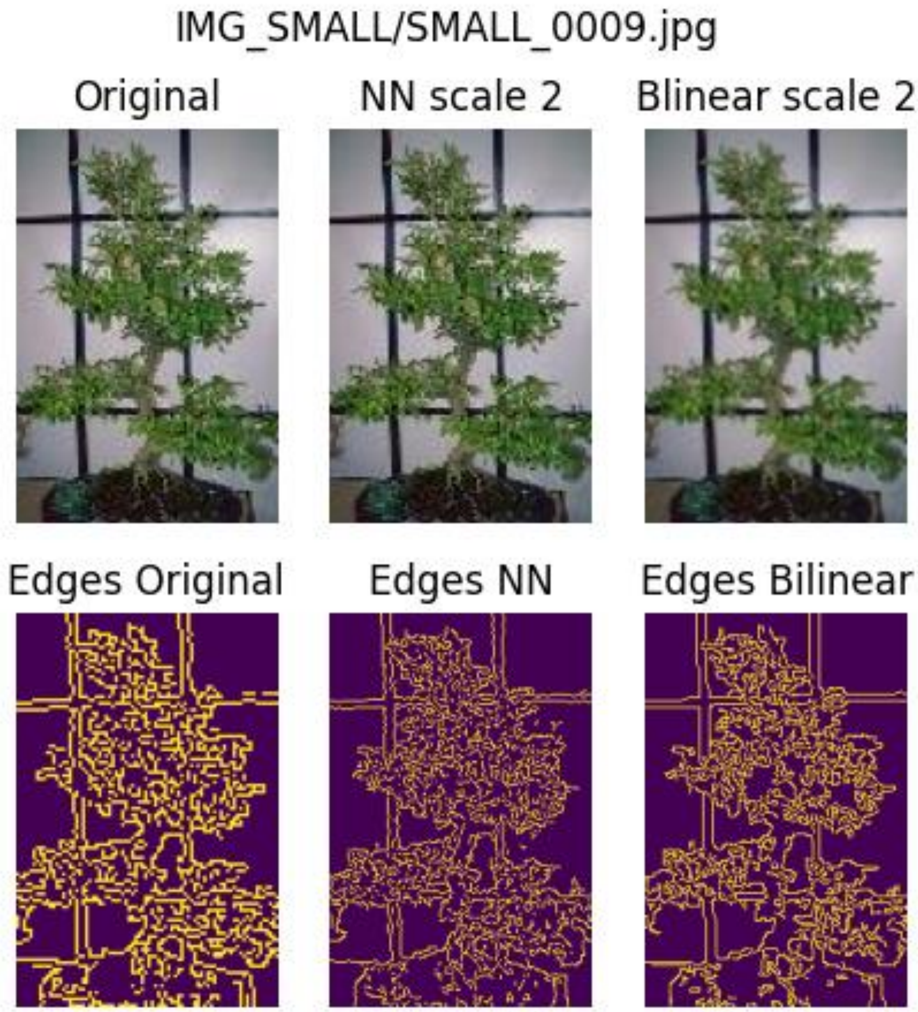
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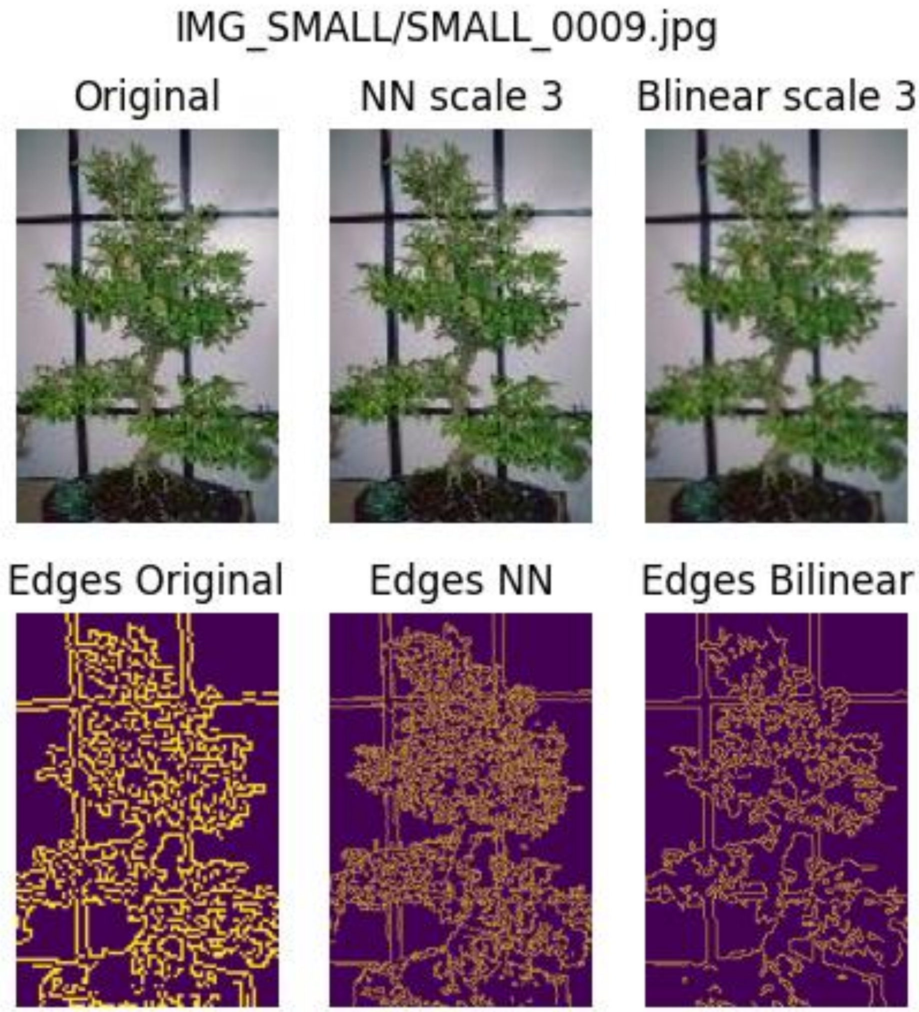
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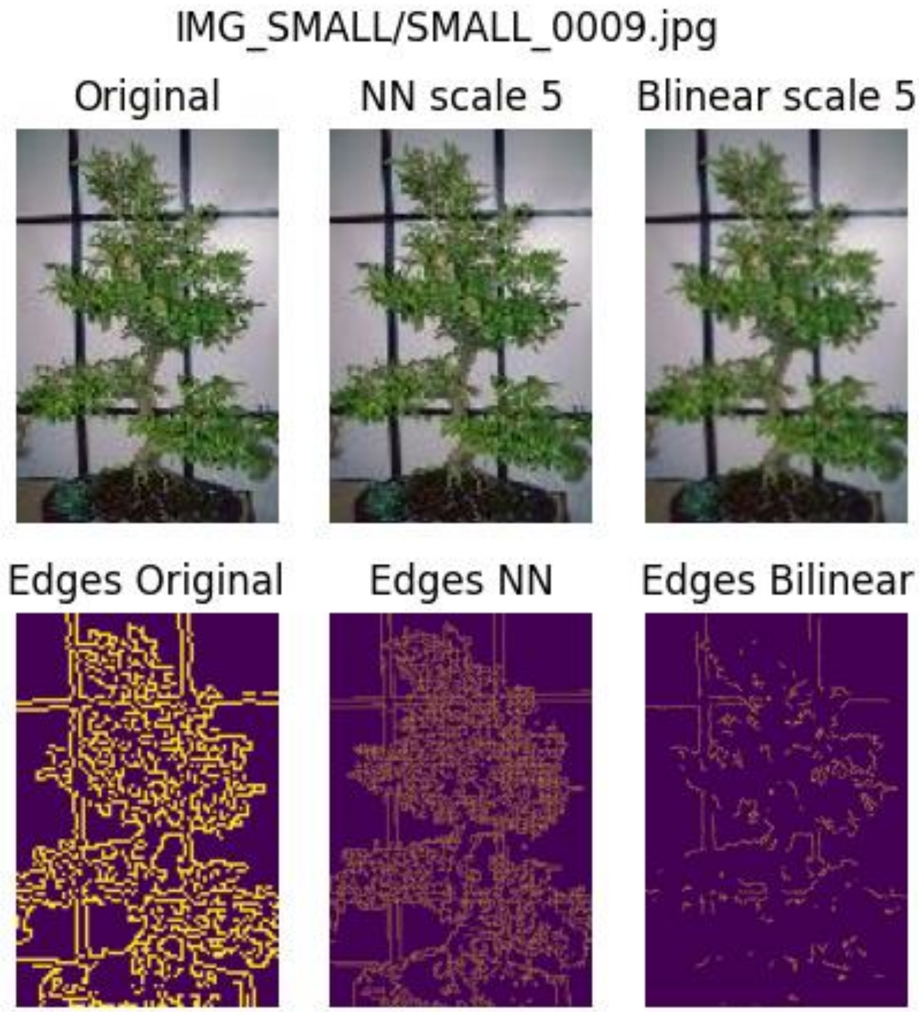
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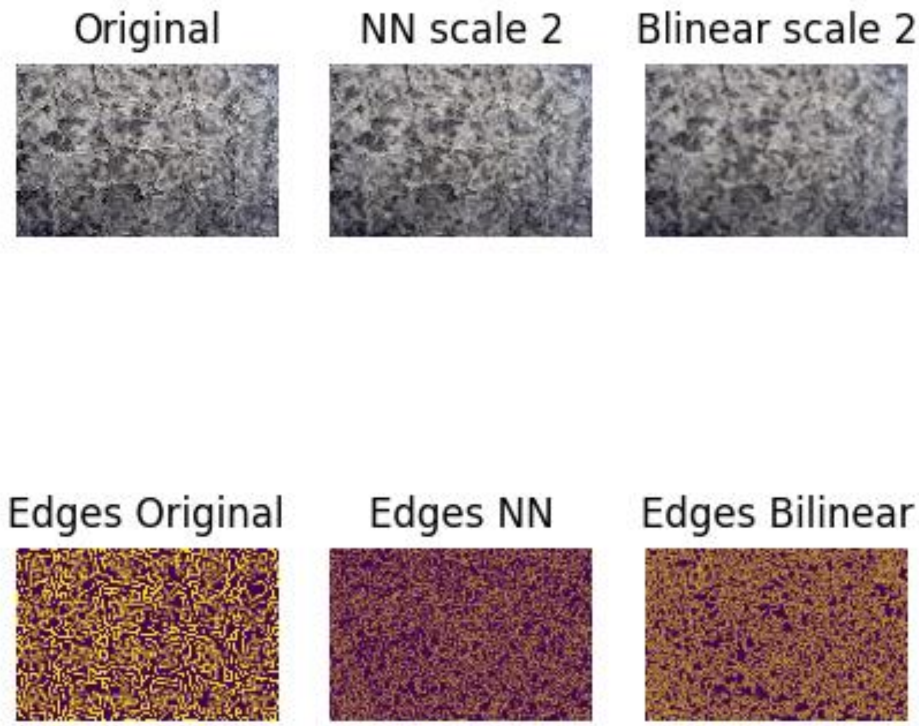

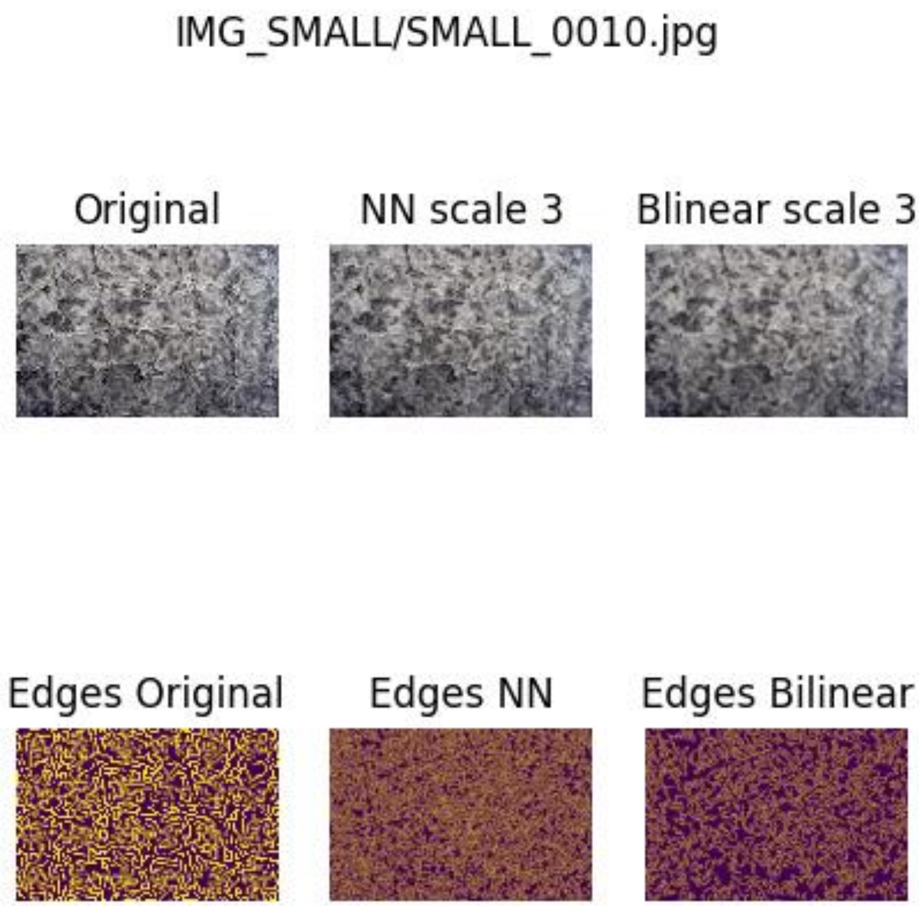
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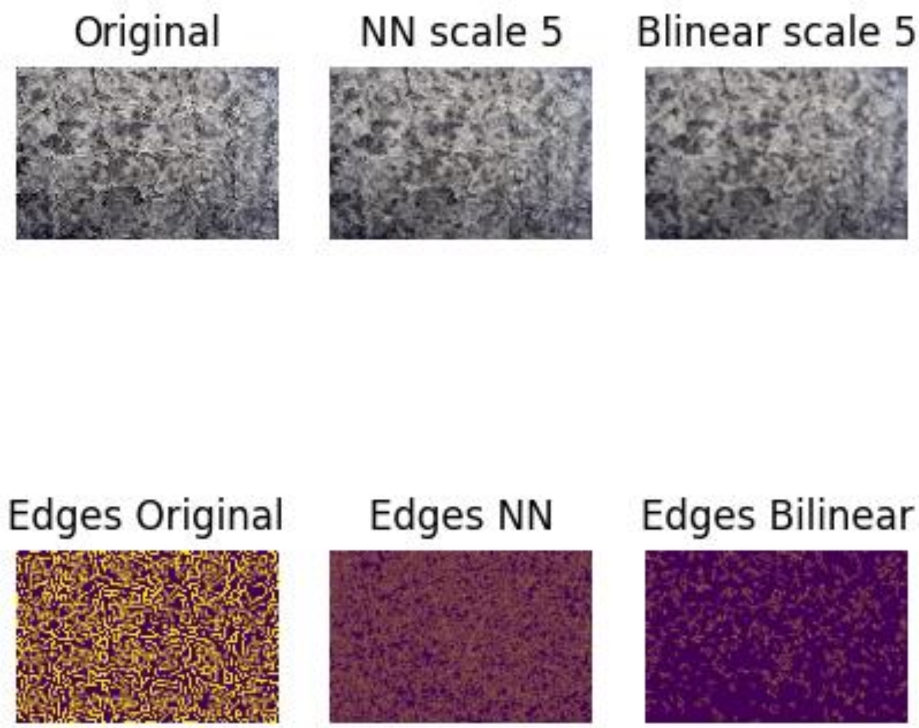
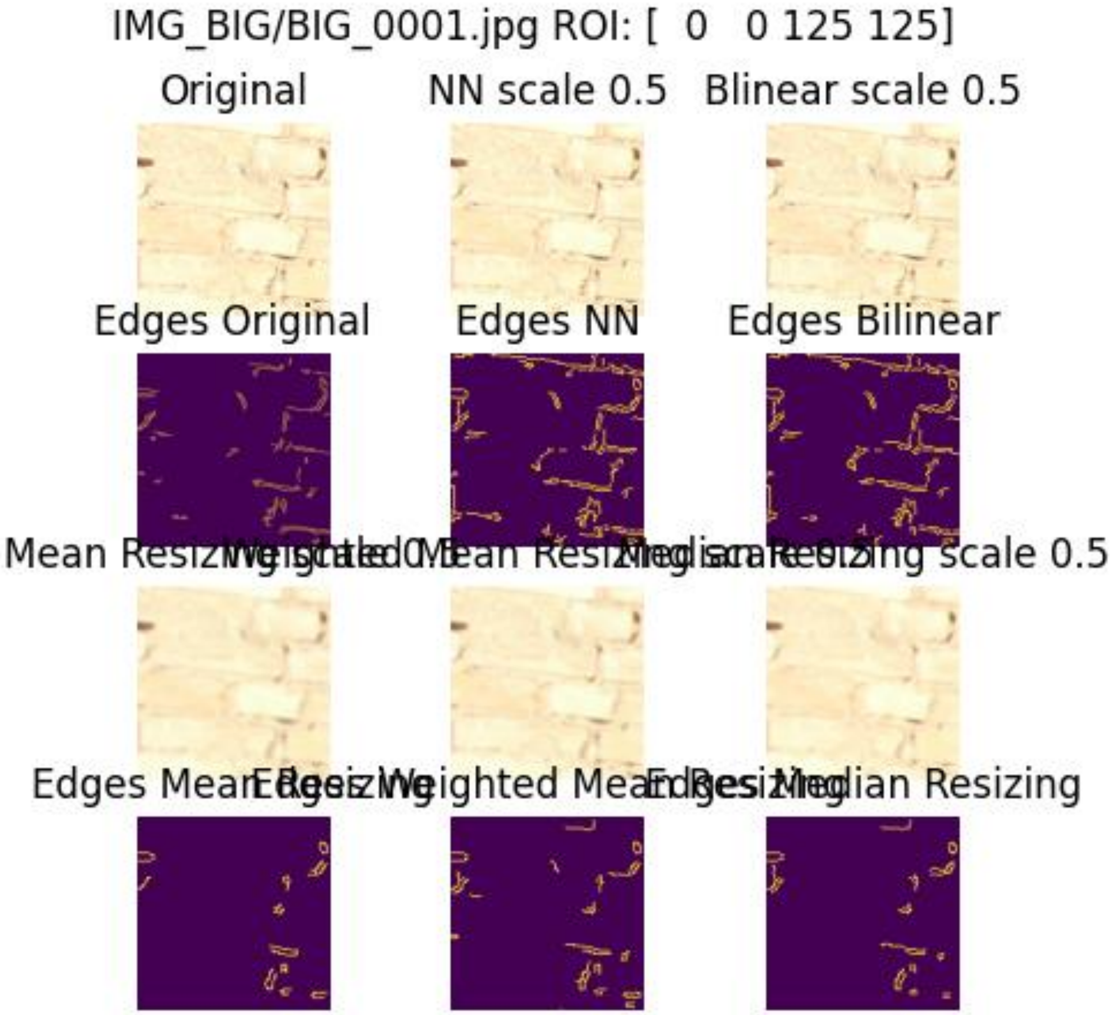
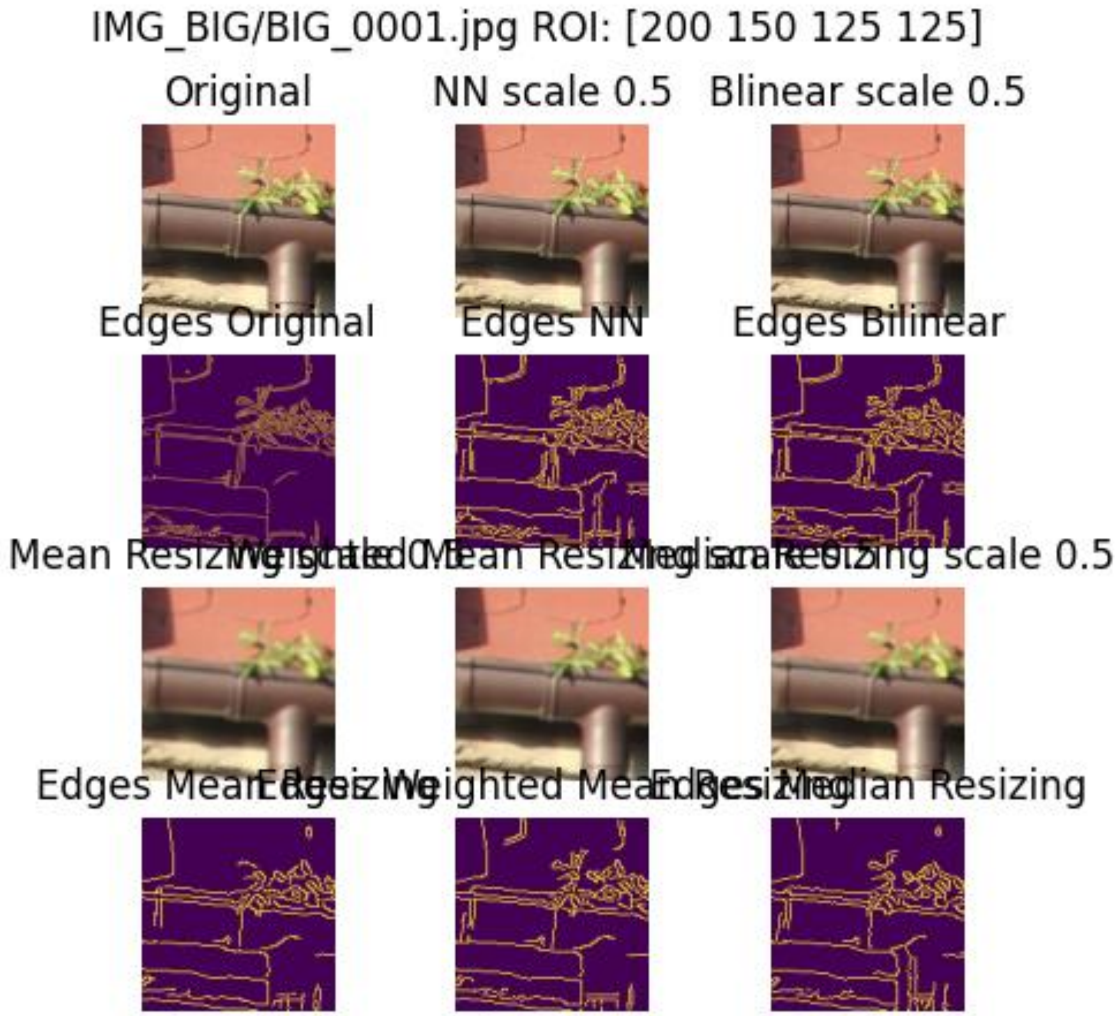


Test algorytmów pomniejszania

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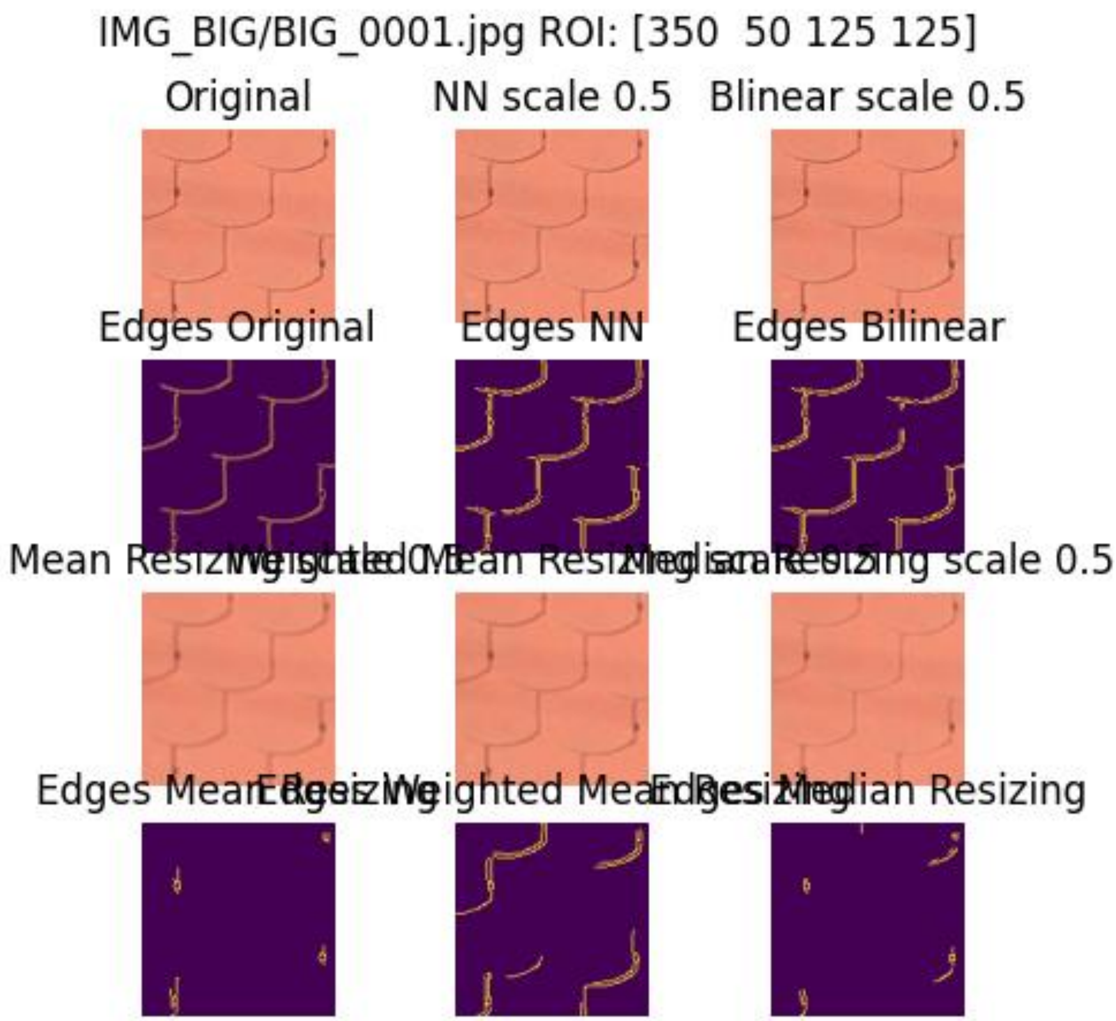
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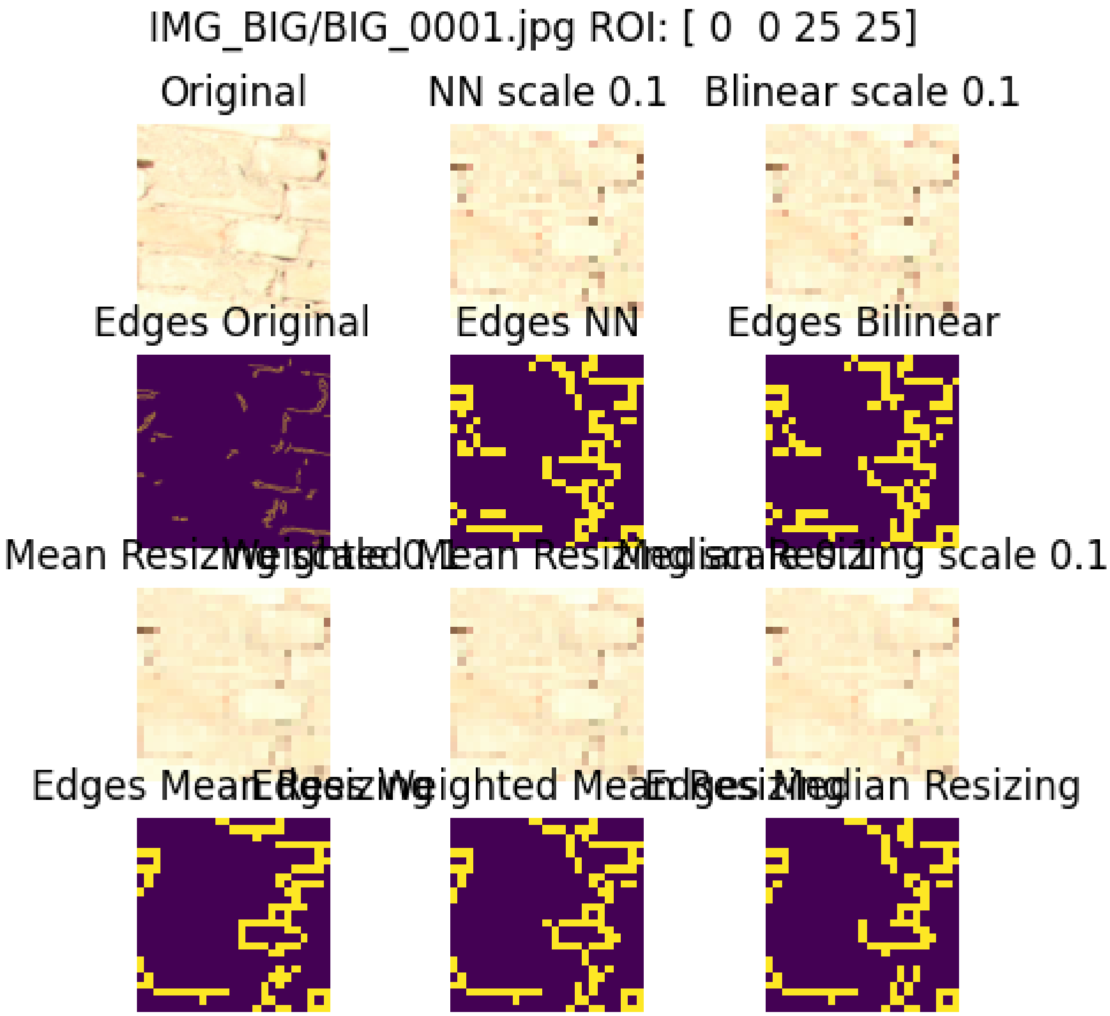
IMG_BIG/BIG_0001.jpg ROI: [0 0 25 25]
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Blinear scale 0.1
Edges Original
Edges NN
Edges Bilinear

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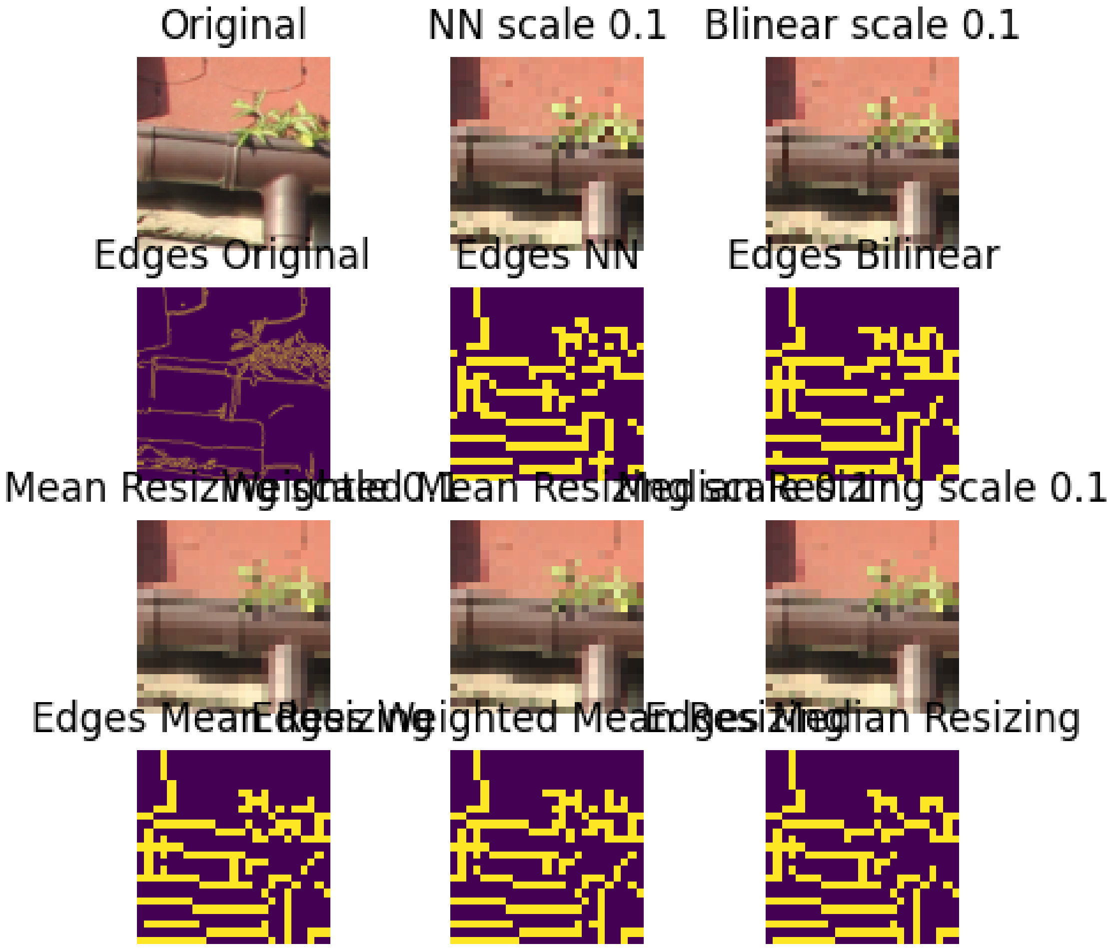

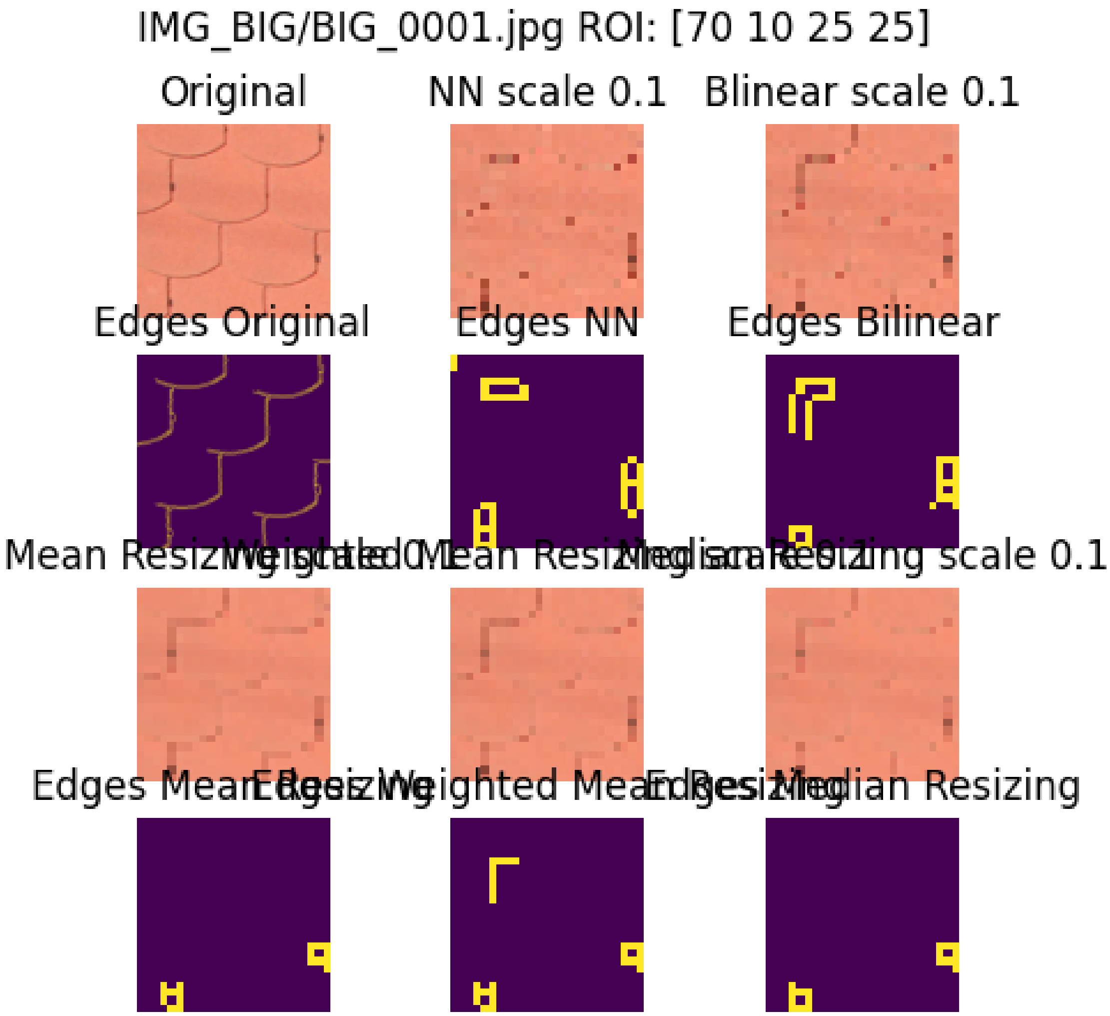
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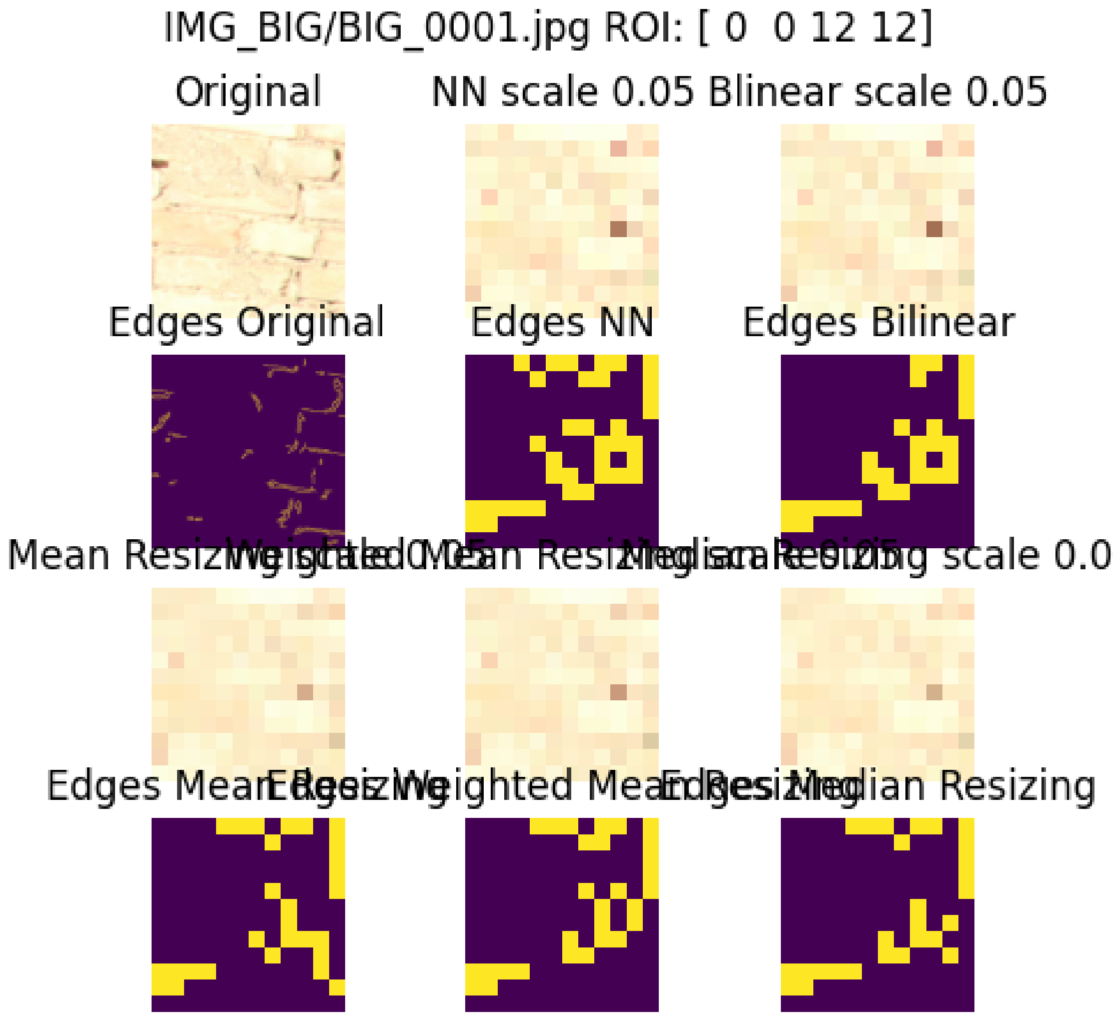
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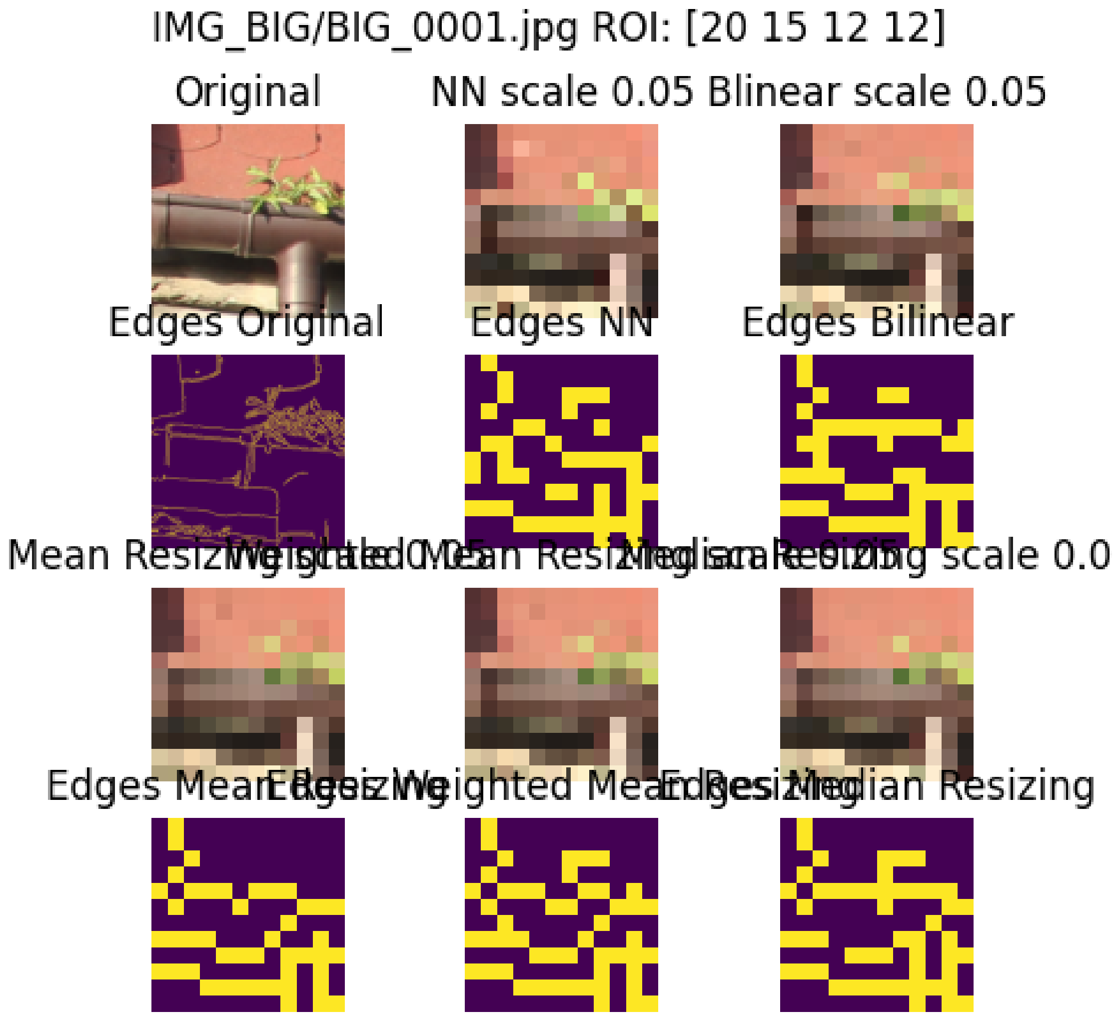
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Edges NN
Edges Bilinear
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Median Resizing scale 0.05
Edges Mean Resizing
Edges Weighted Mean Resizing
Edges Median Resizing


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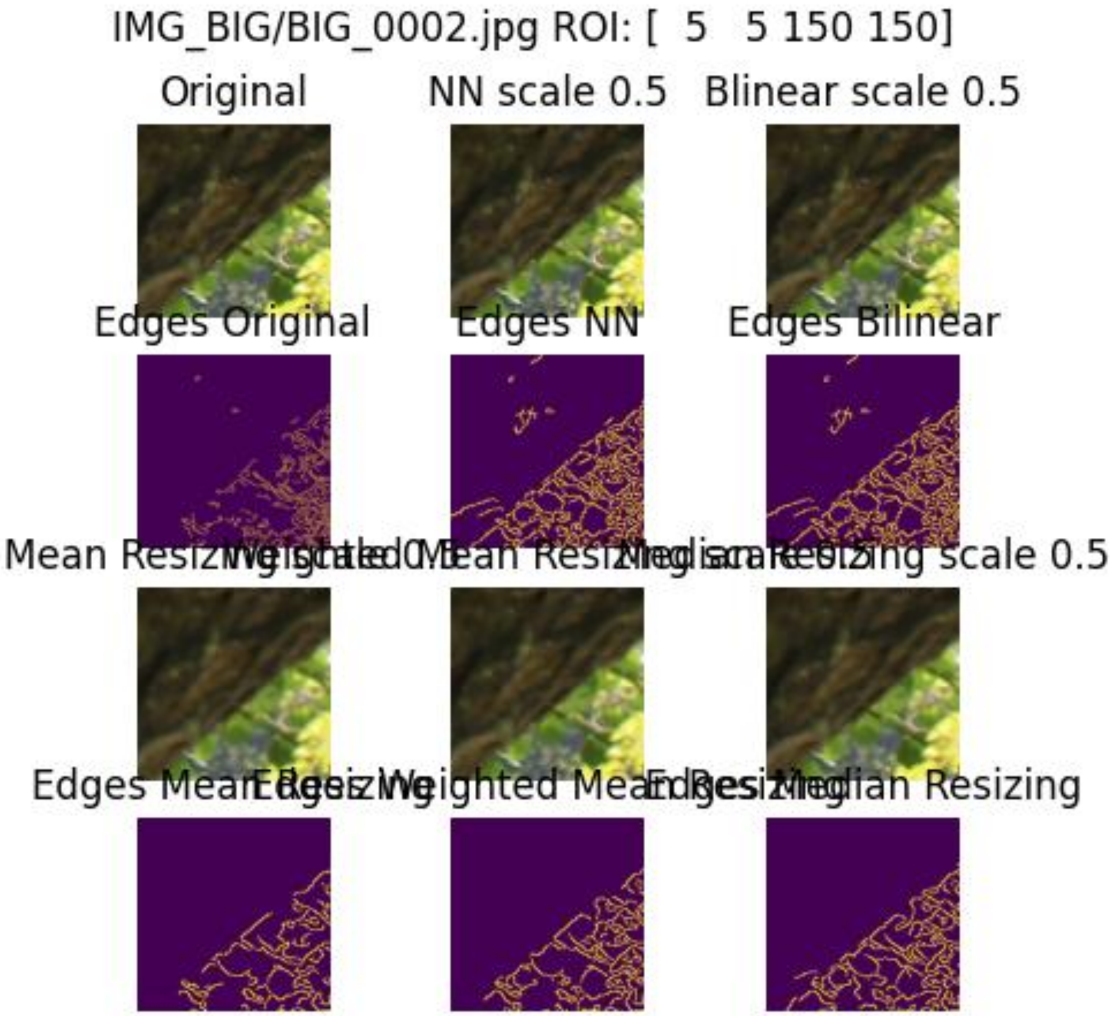
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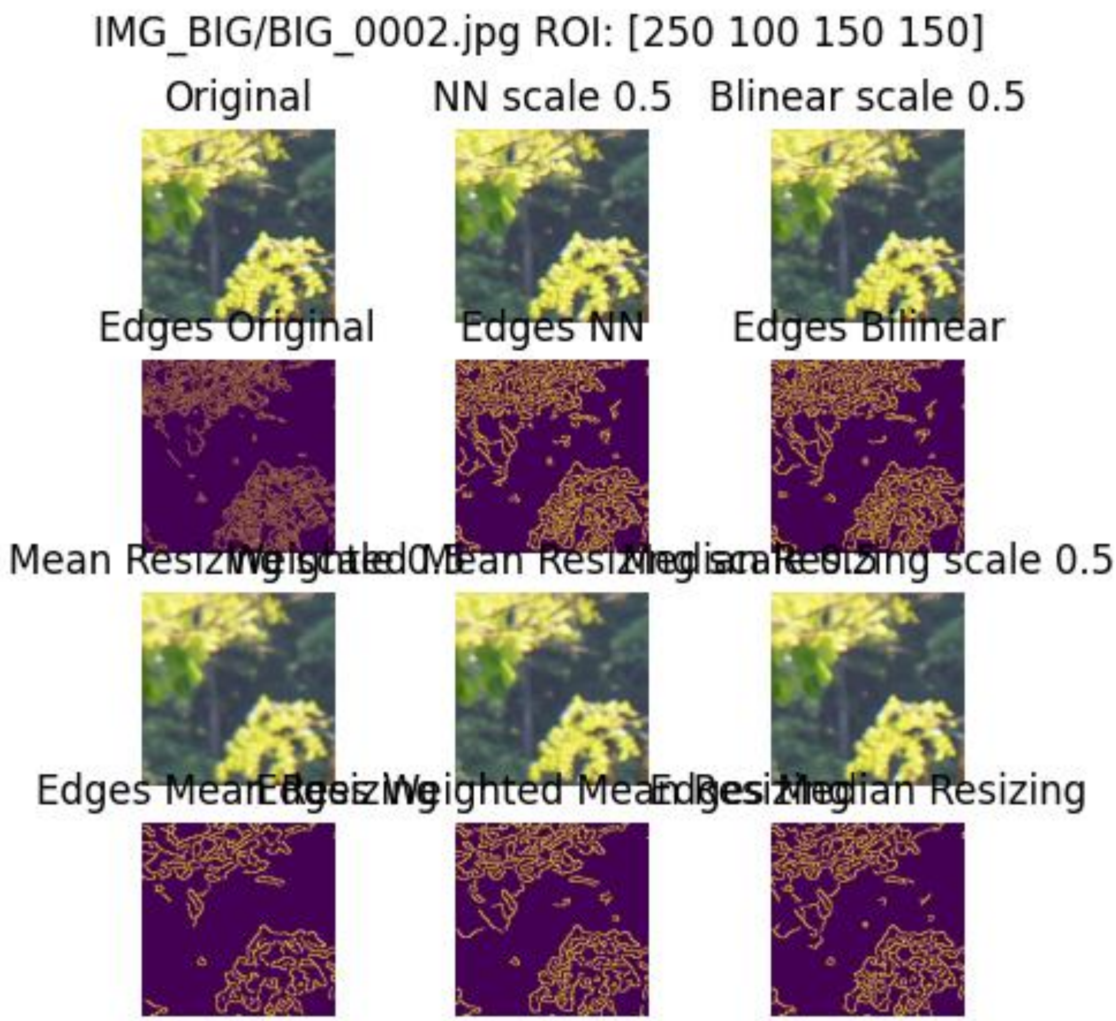

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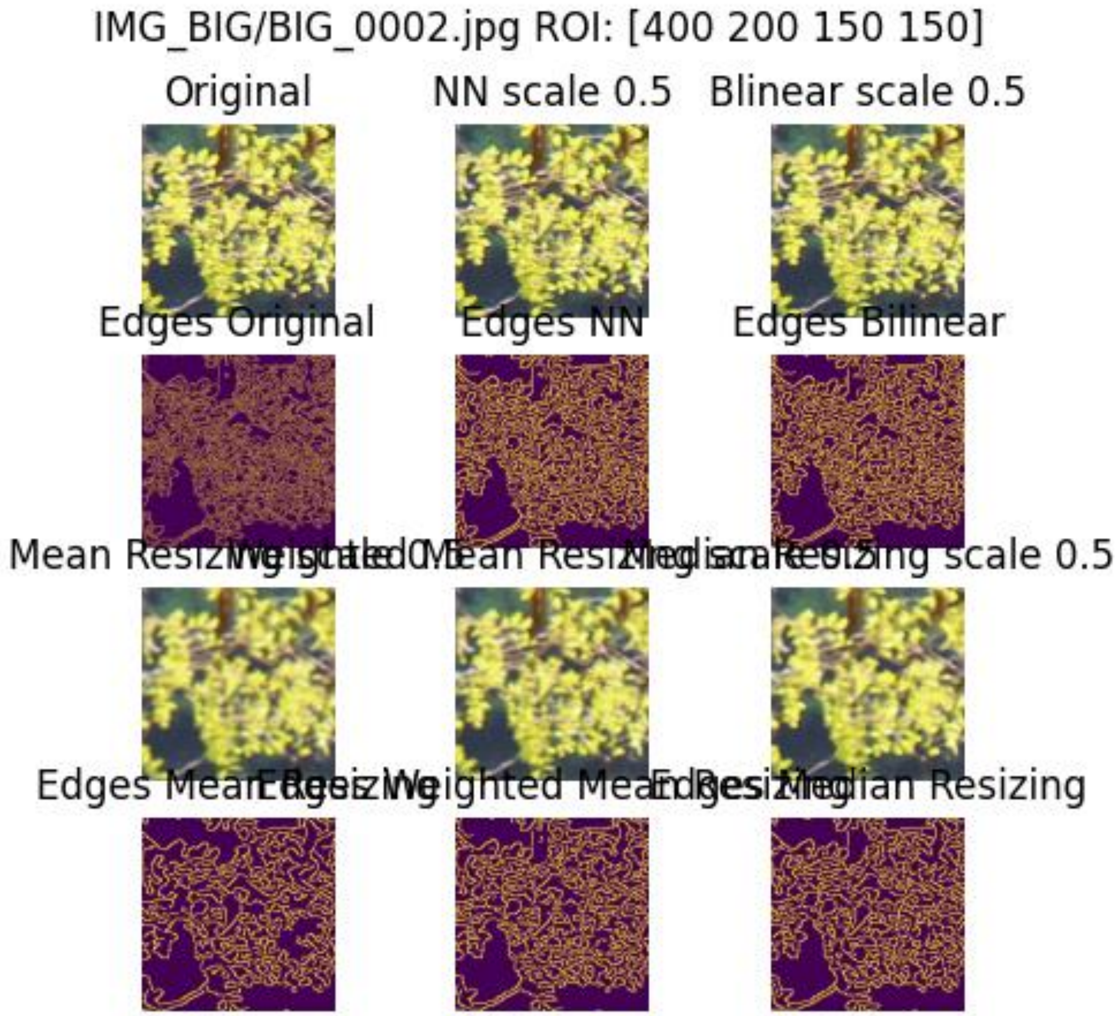
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Mean Resi
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Edges Mean
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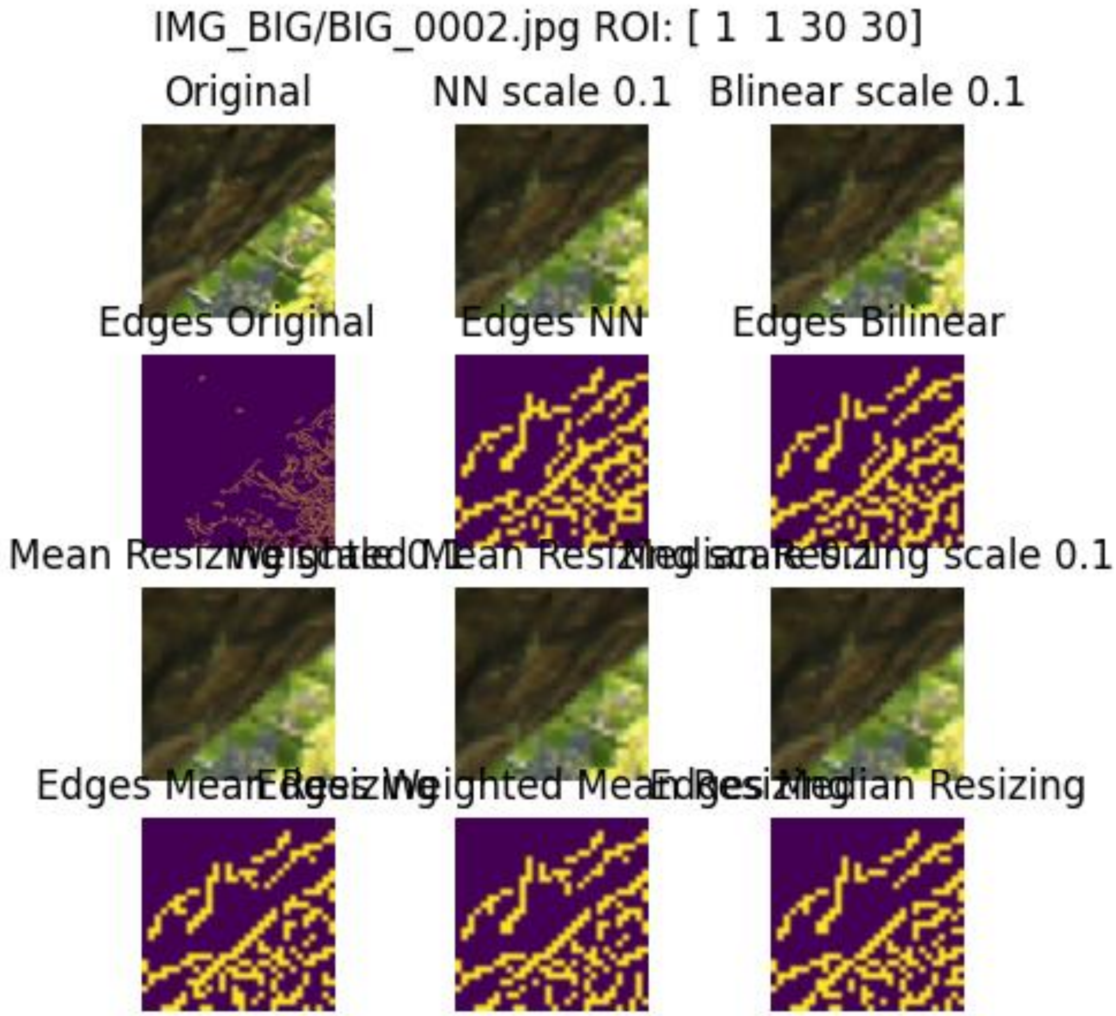
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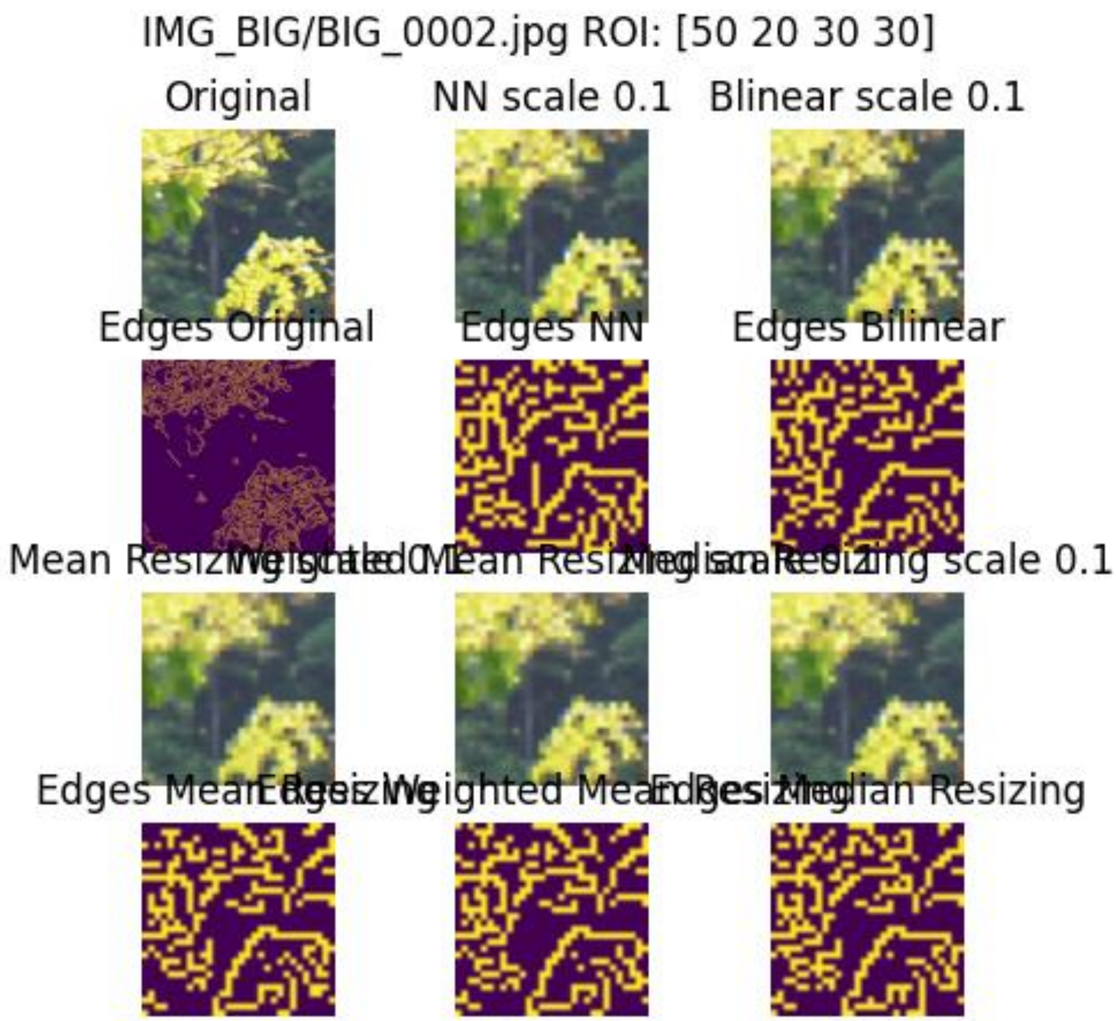

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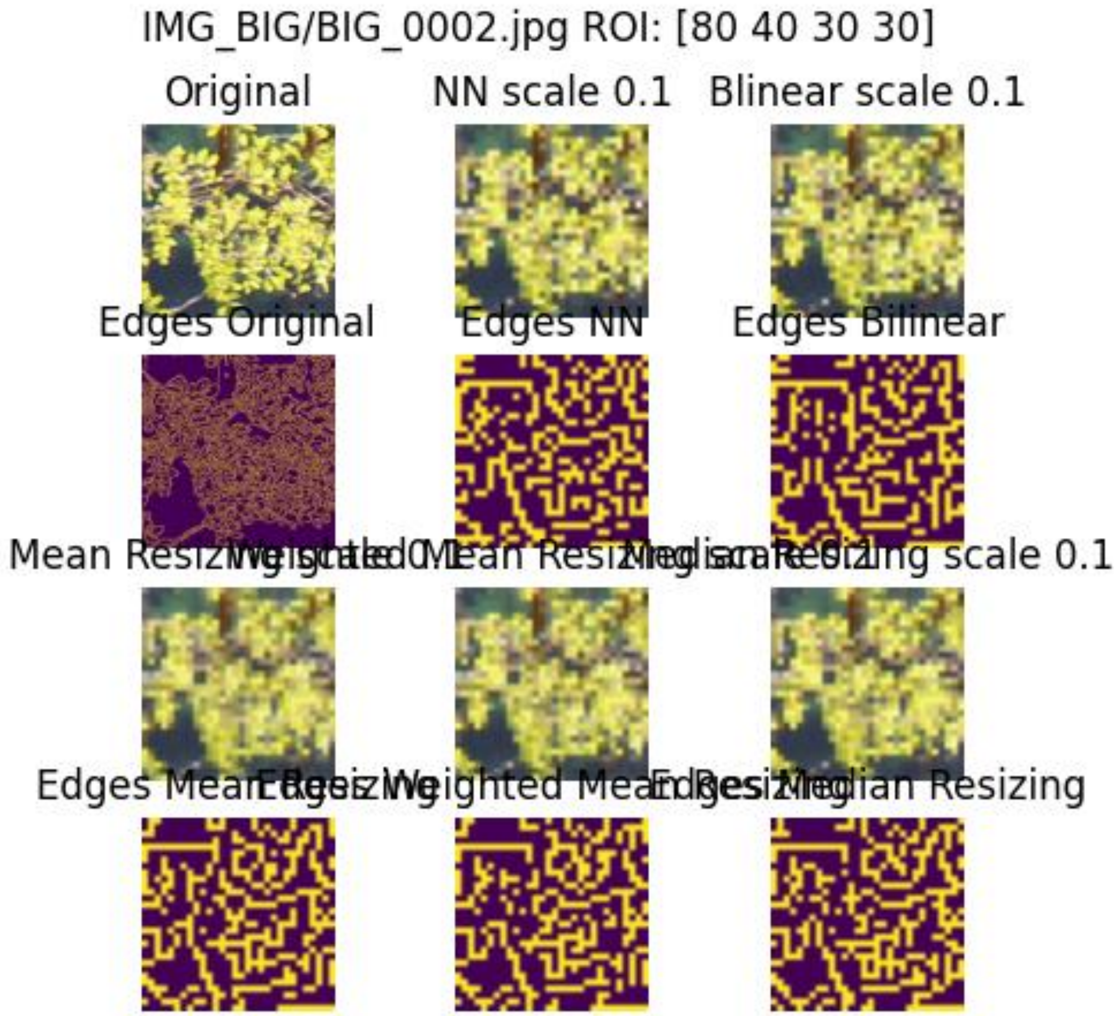
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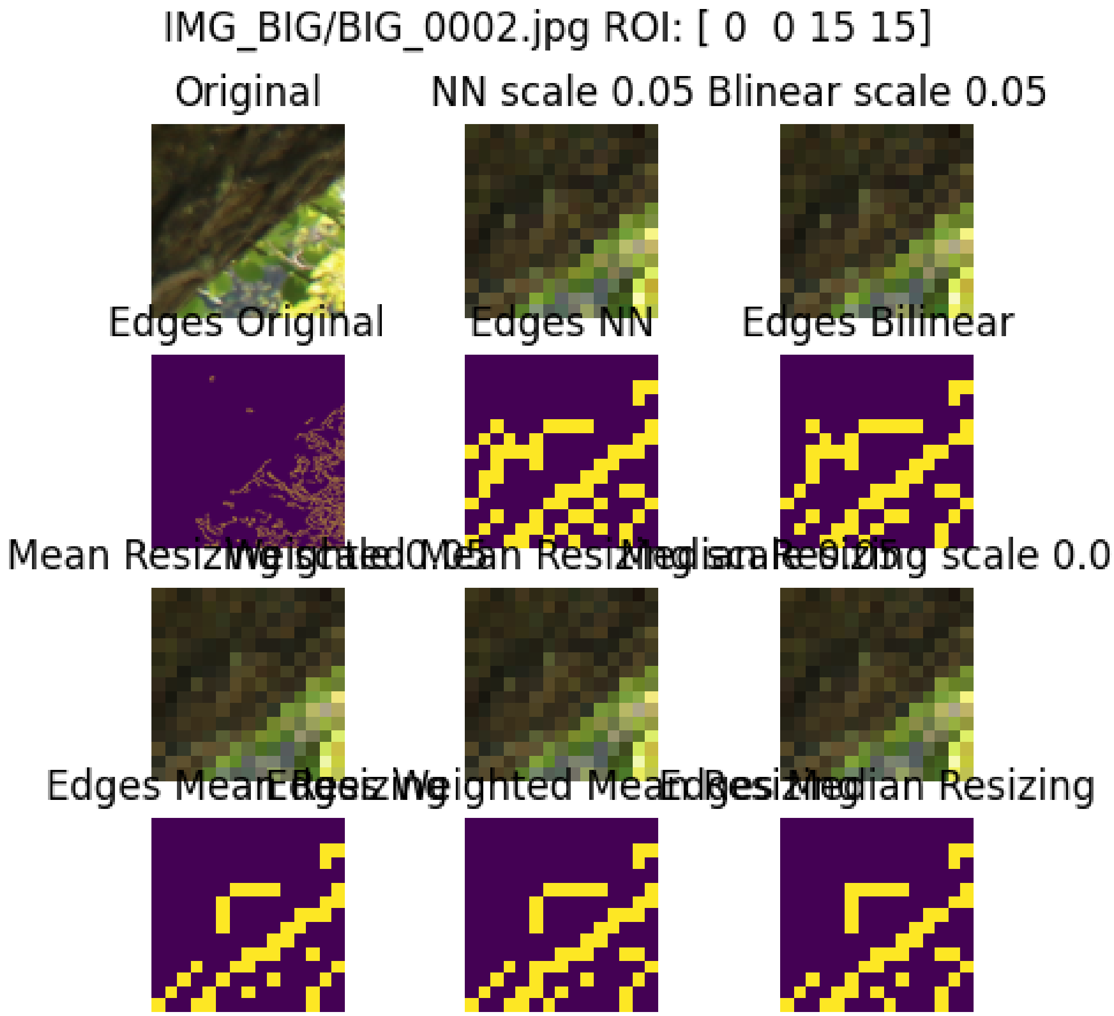
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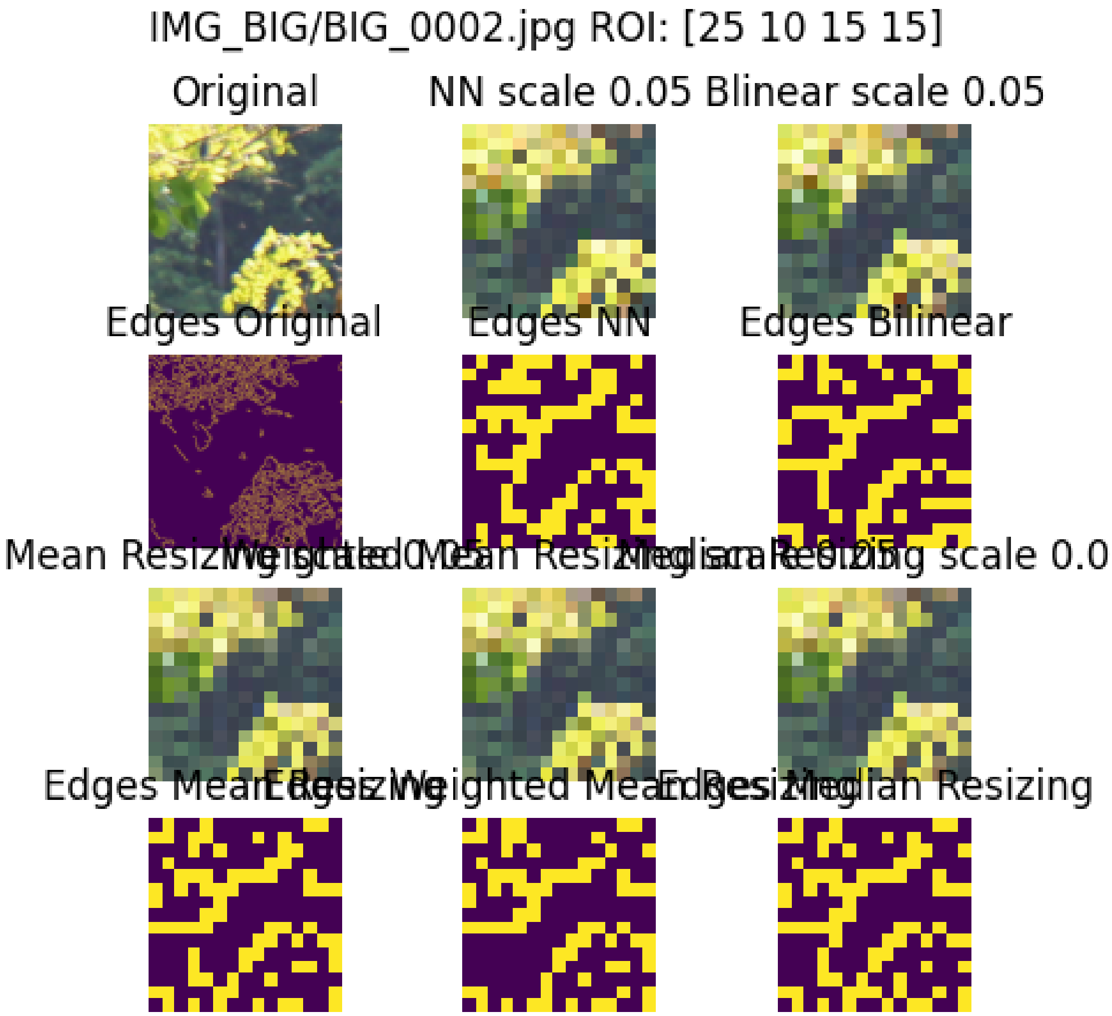
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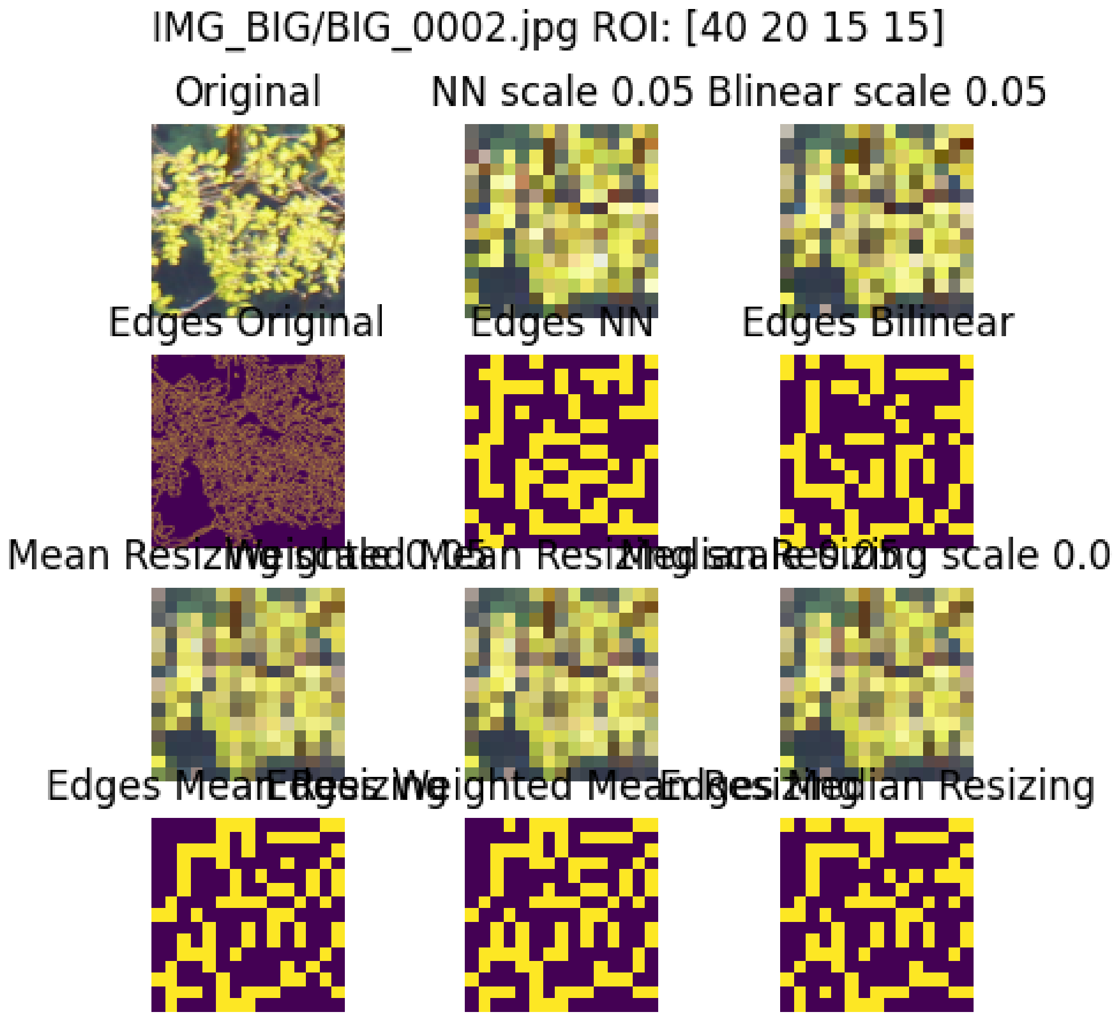
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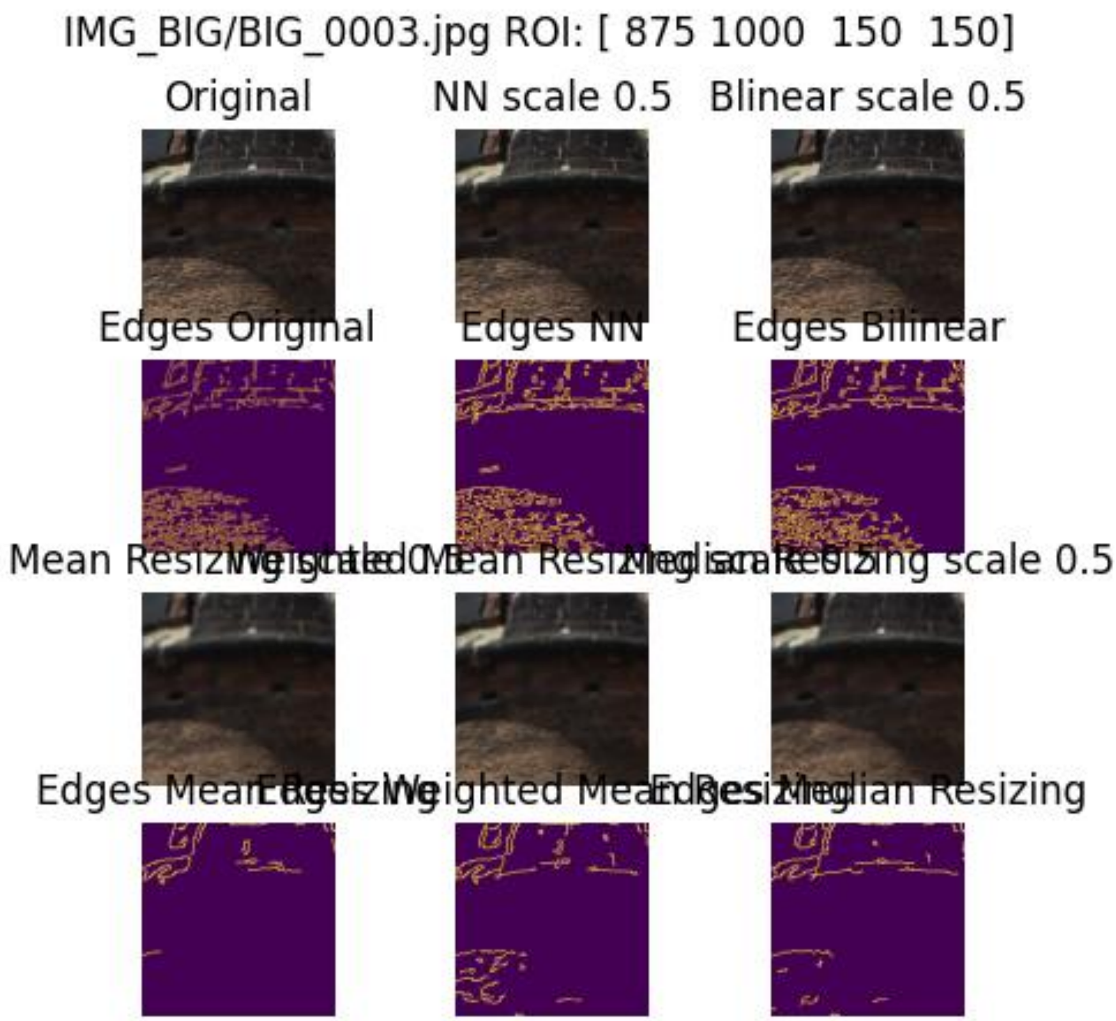
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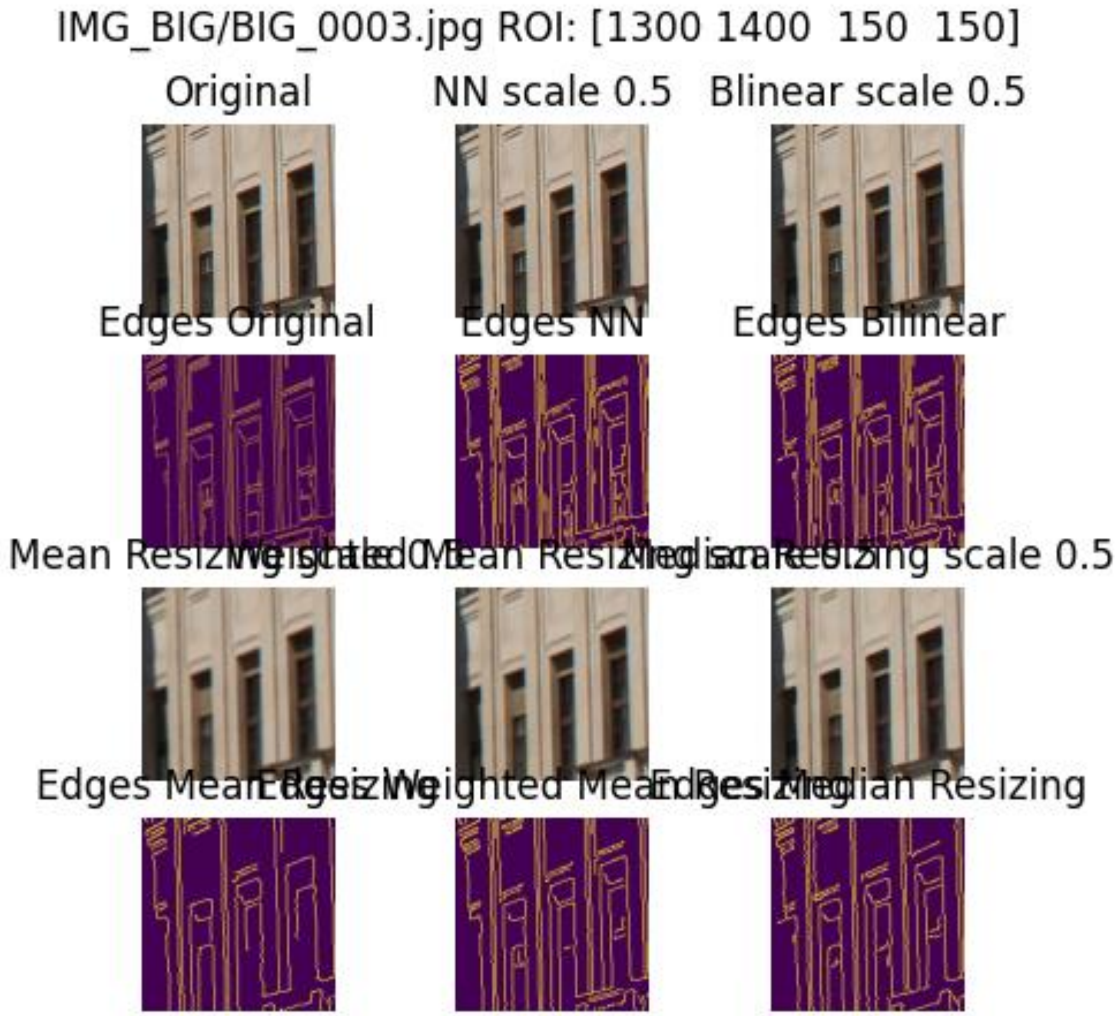
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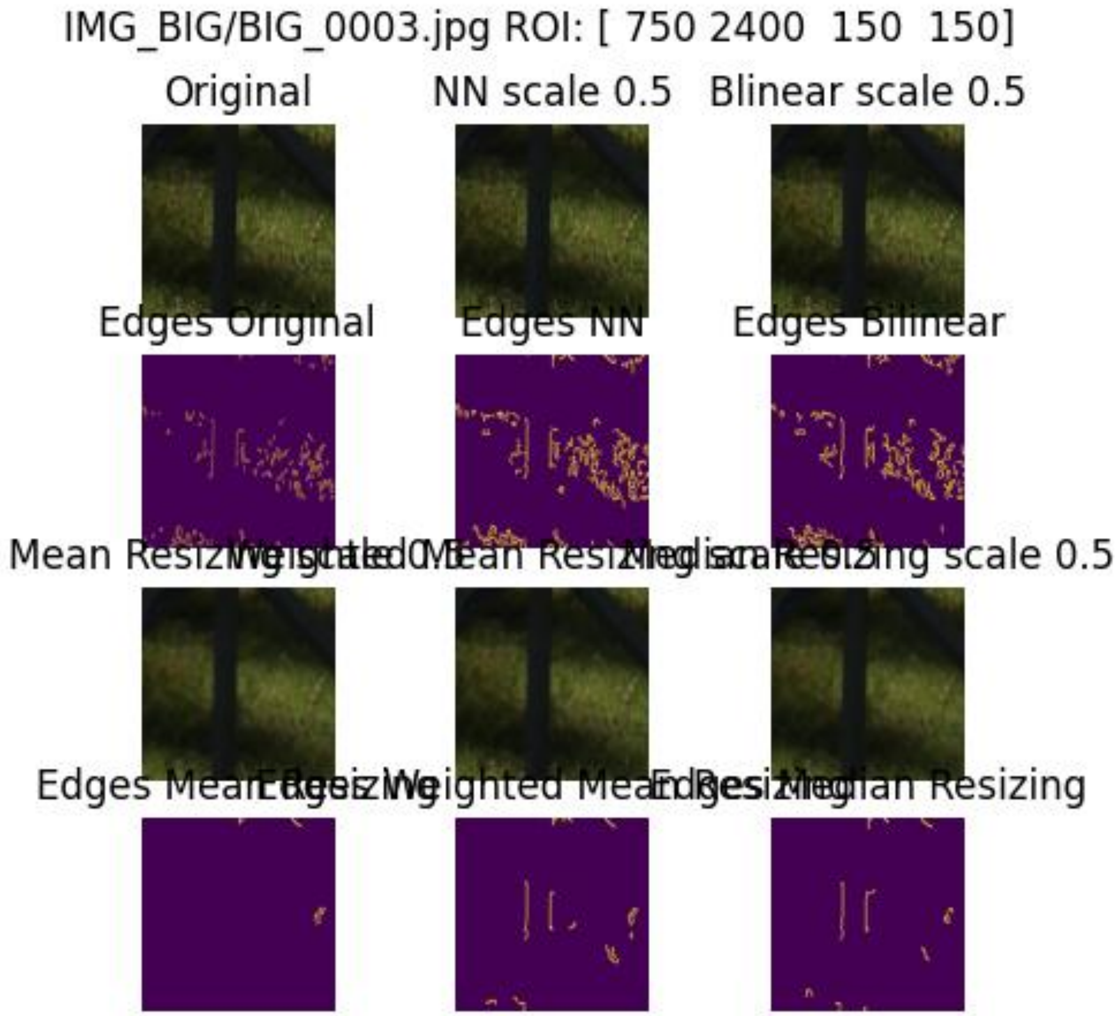
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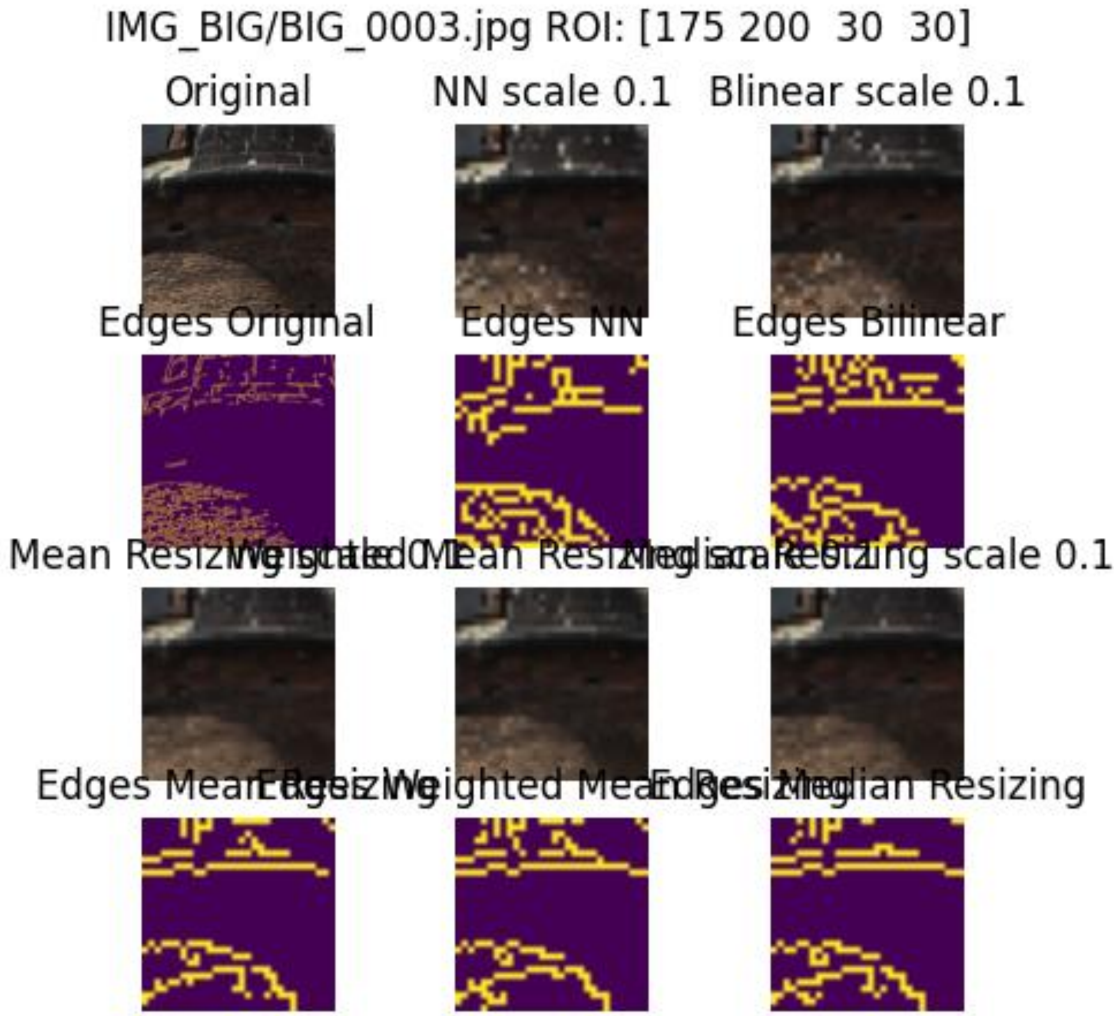
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Blinear scale 0.1
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Edges NN
Edges Bilinear


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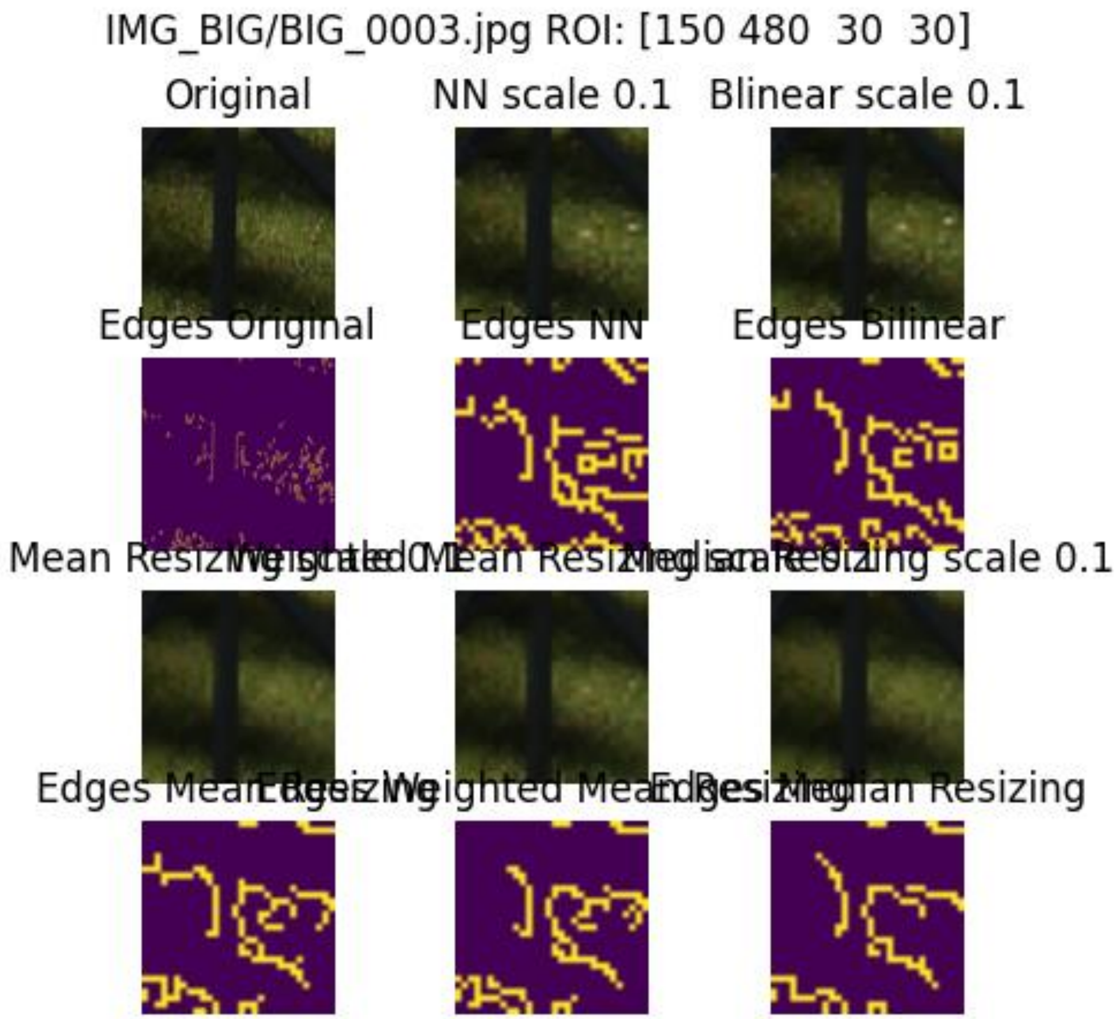
Edges Median Resizing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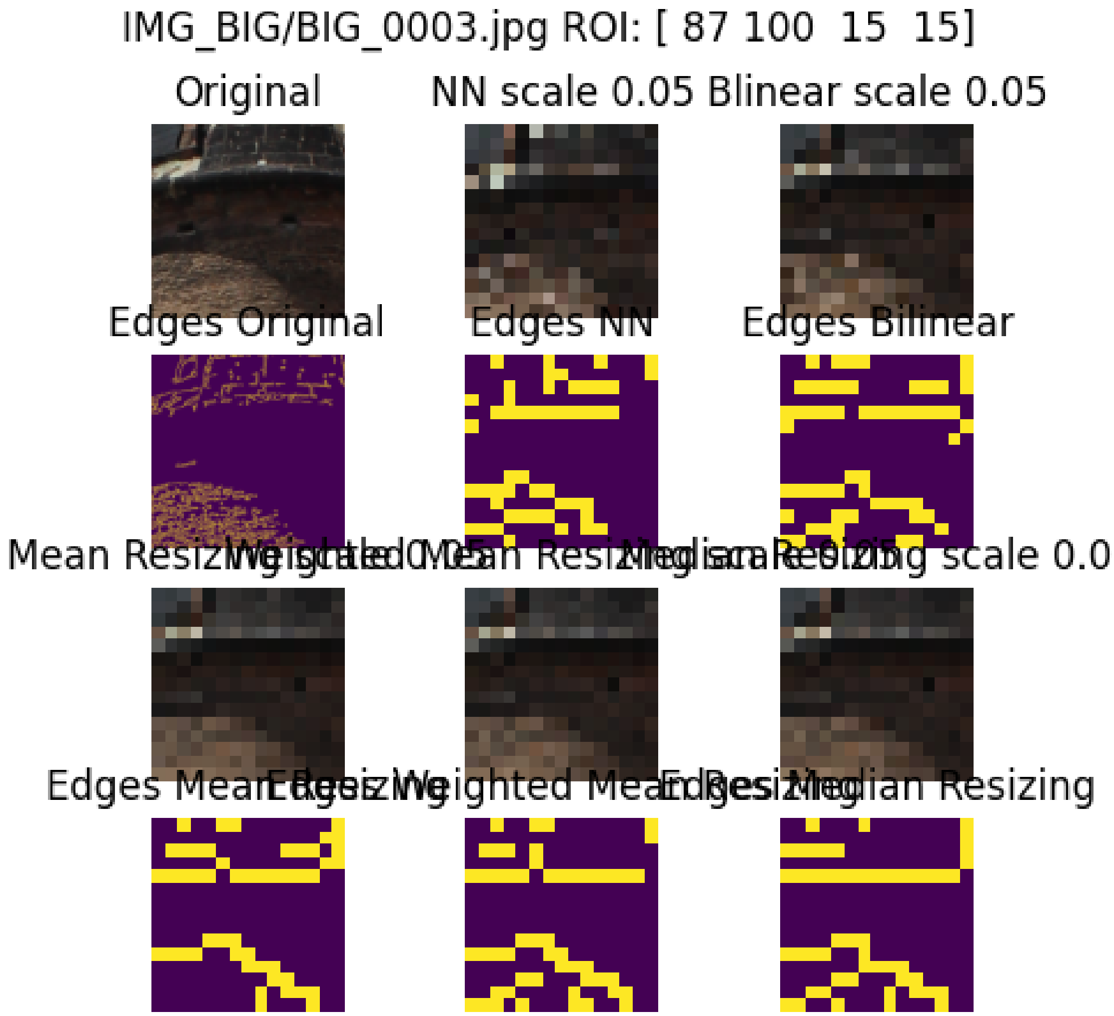
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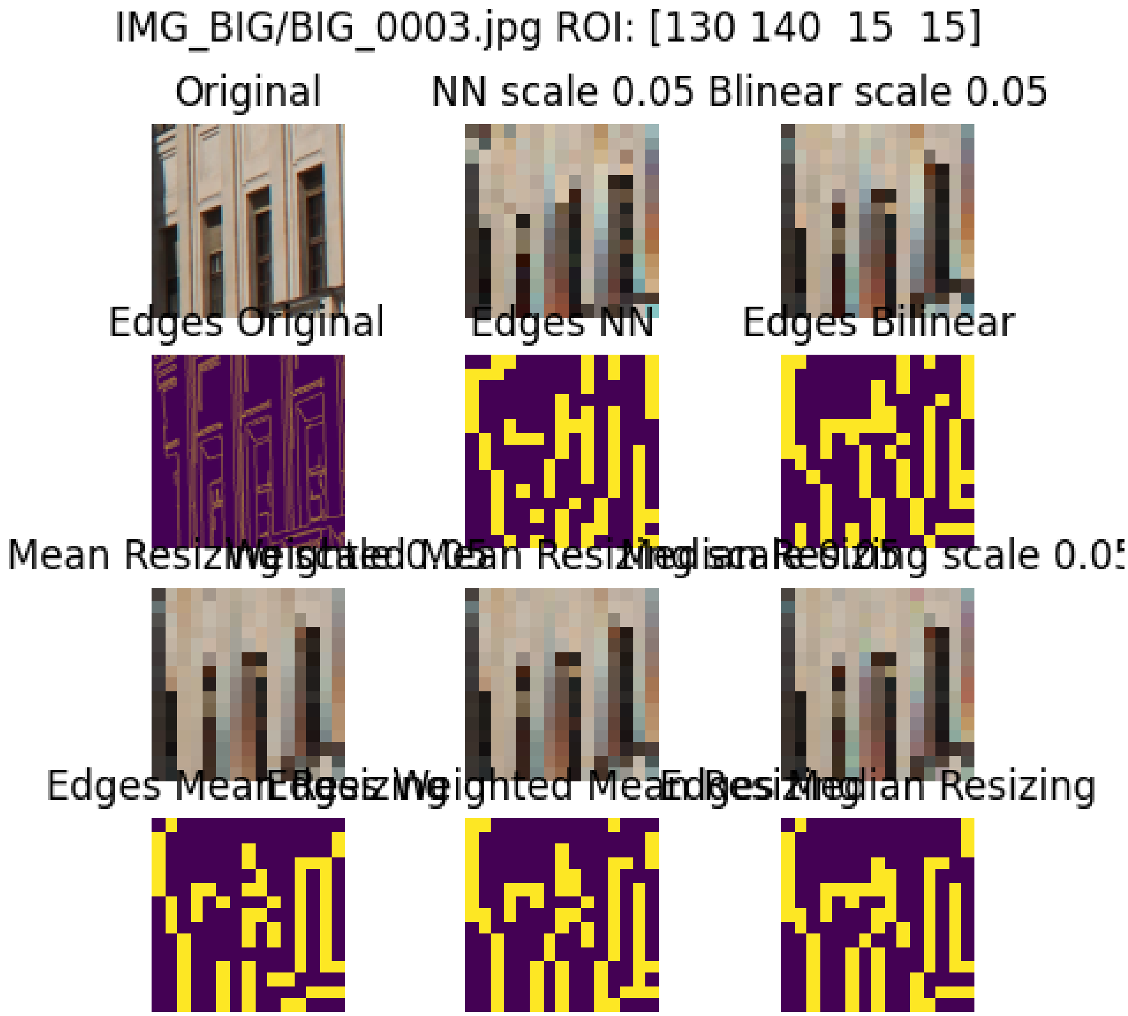
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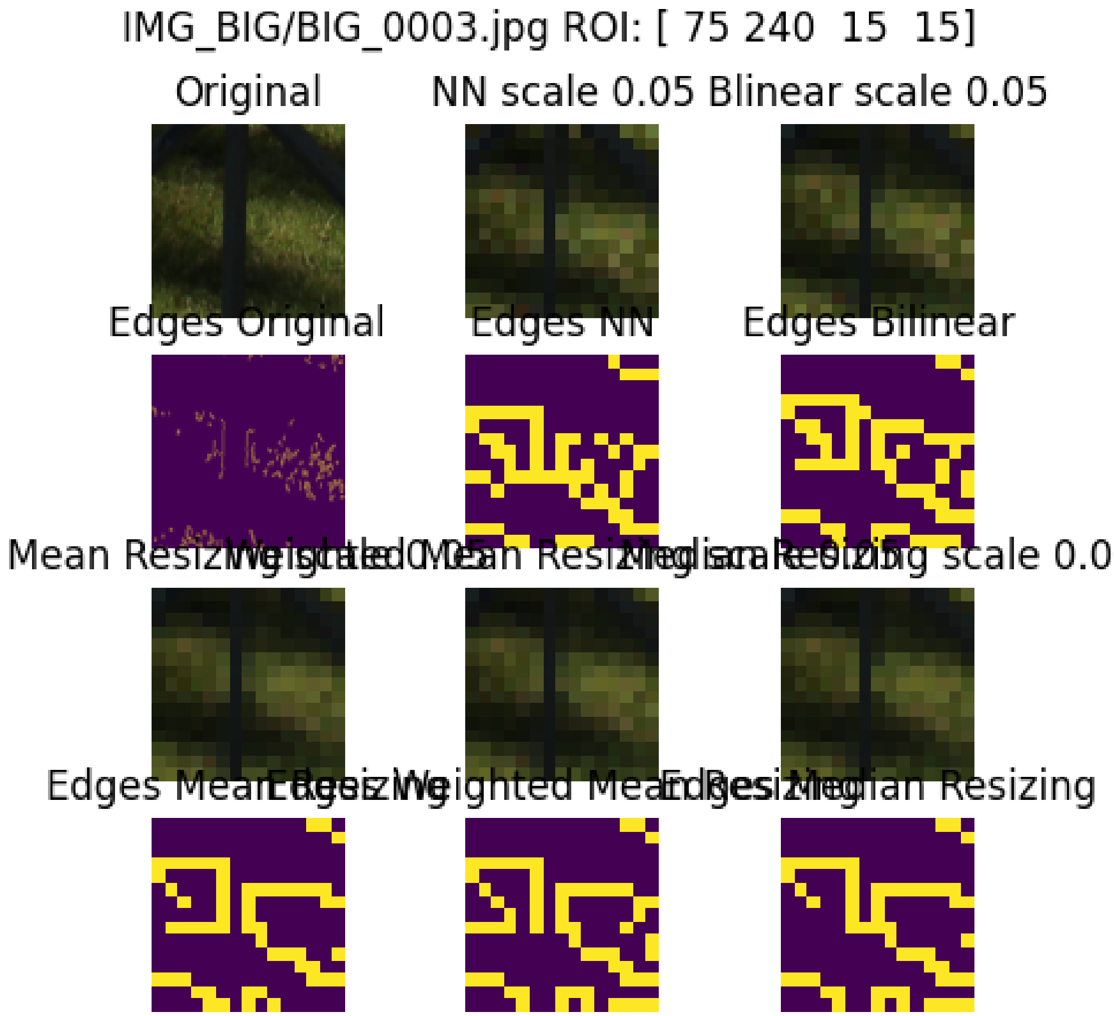
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Edges Bilinear
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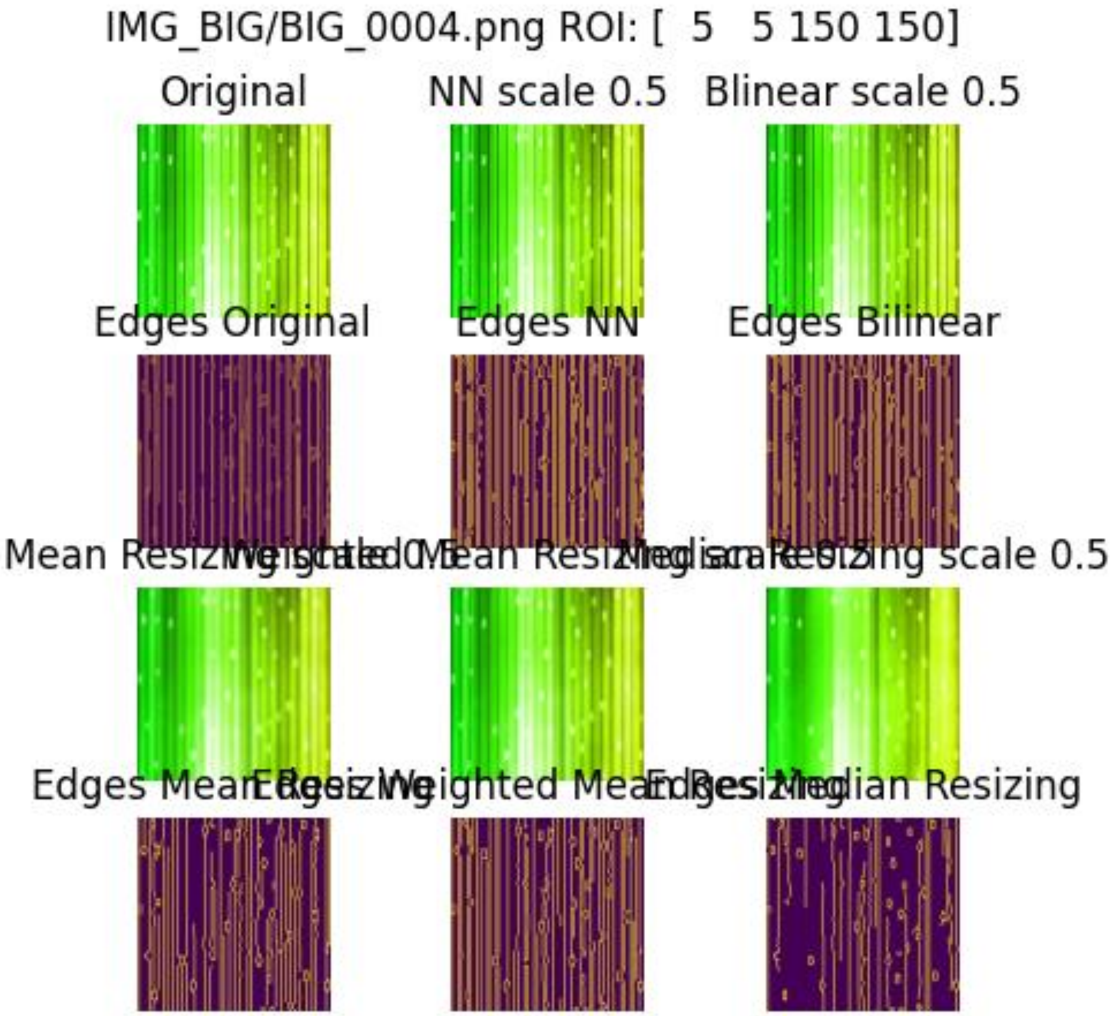
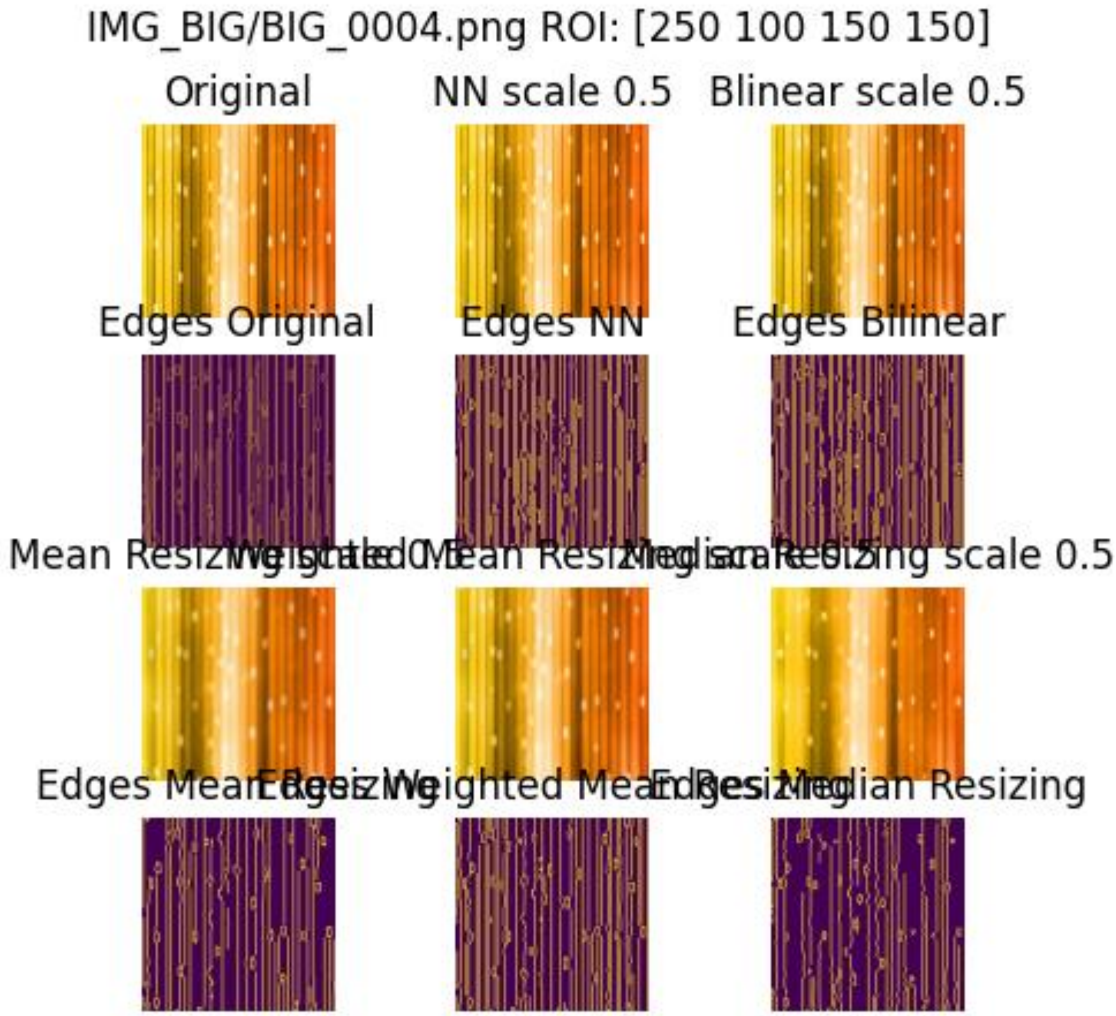
Weighted Mean Resizing scale 0.05
Median Resizing scale 0.05
Edges Mean Resizing
Edges Weighted Mean Resizing
Edges Median Resizing


IMG_BIG/BIG_0004.png ROI: [5 5 150 150]
Original
NN scale 0.5
Blinear scale 0.5
Edges Original
Edges NN
Edges Bilinear
Mean Resizing scale 0.5
Weighted Mean Resizing scale 0.5
Median Resizing scale 0.5
Edges Mean Resizing
Edges Weighted Mean Resizing
Edges Median Resizing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Blinear scale 0.5
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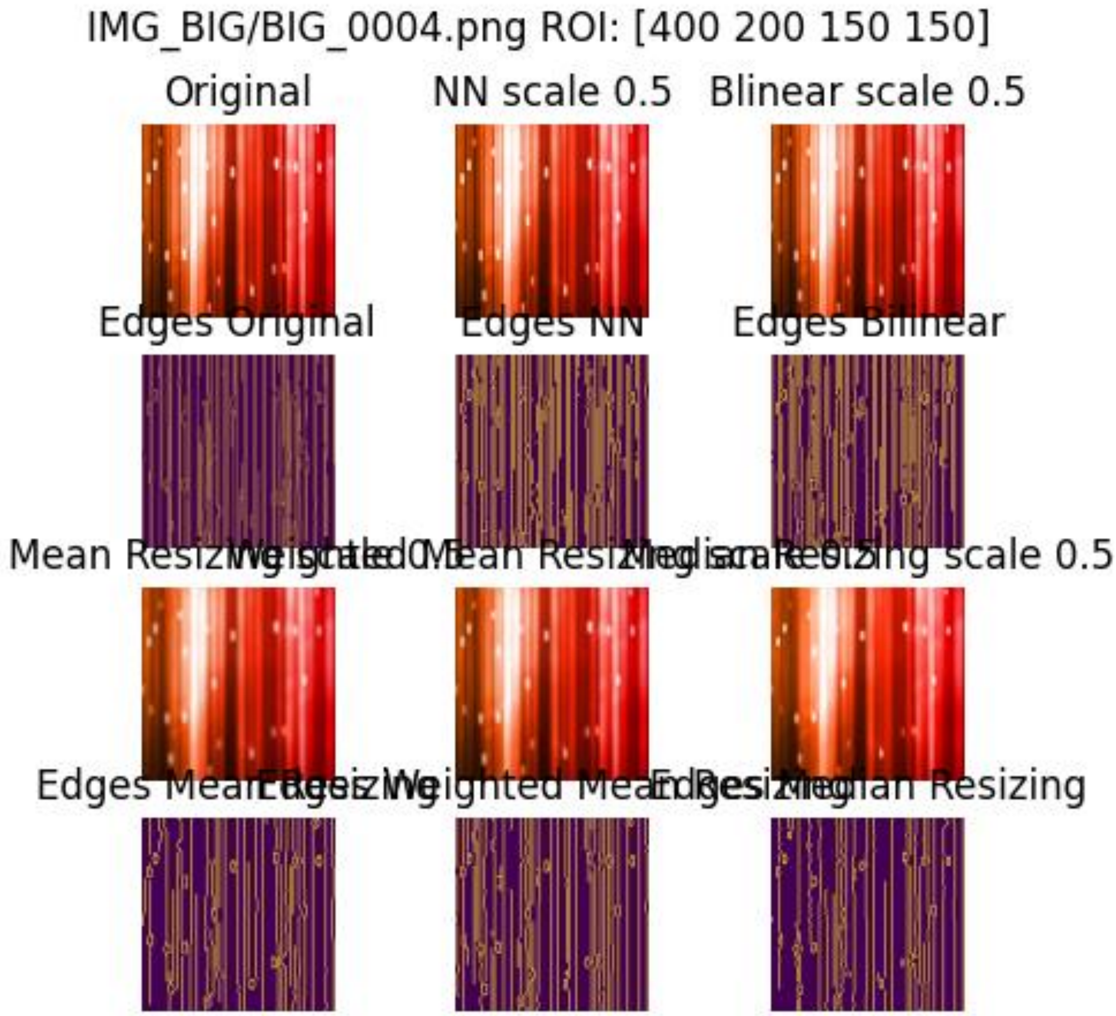
Edges NN
Edges Bilinear


IMG_BIG/BIG_0004.png ROI: [400 200 150 150]
Original
NN scale 0.5
Blinear scale 0.5
Edges Original
Edges NN
Edges Bilinear
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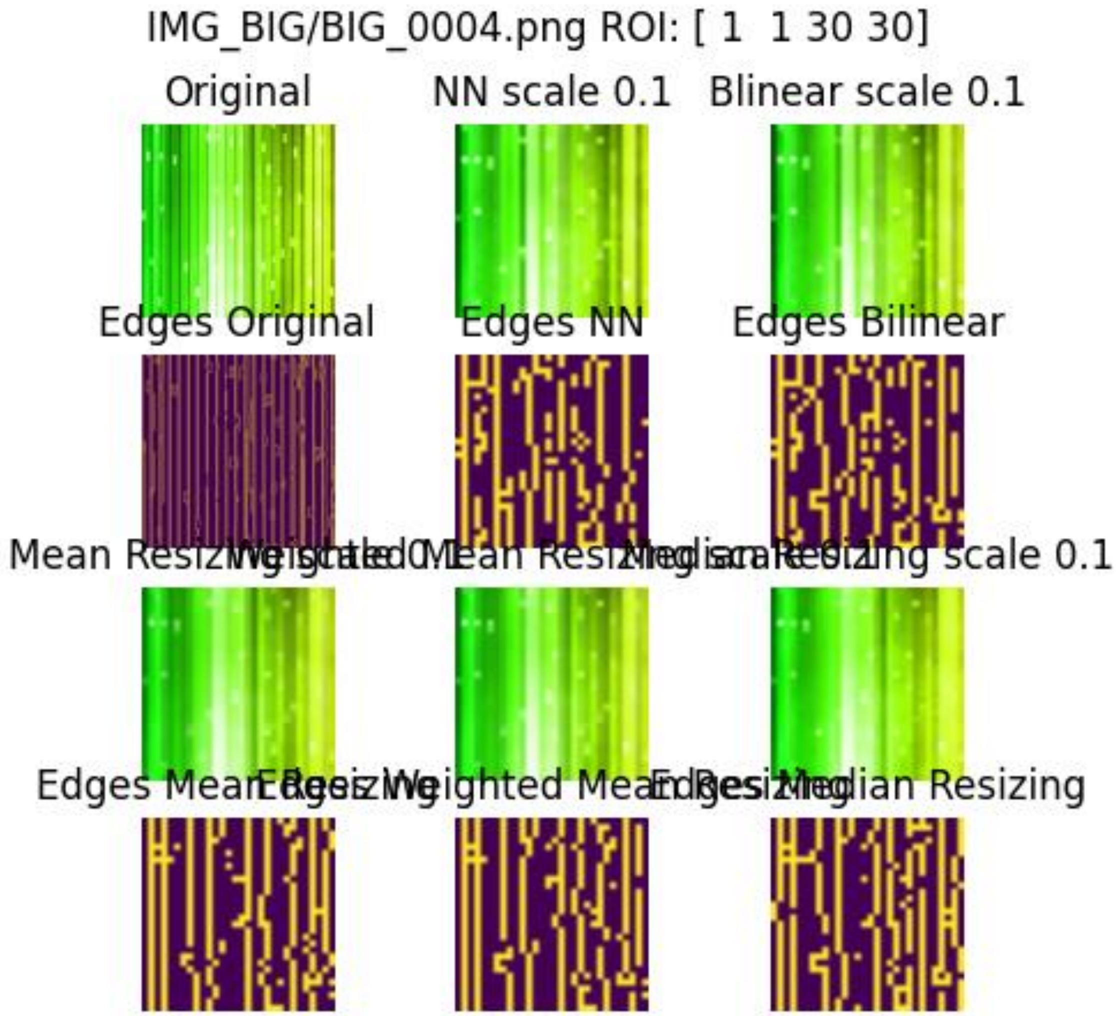
Weighted Mean Resizing scale 0.5
Median Resizing scale 0.5
Edges Mean Resizing
Edges Weighted Mean Resizing
Edges Median Resizing


IMG_BIG/BIG_0004.png ROI: [1 1 30 30]
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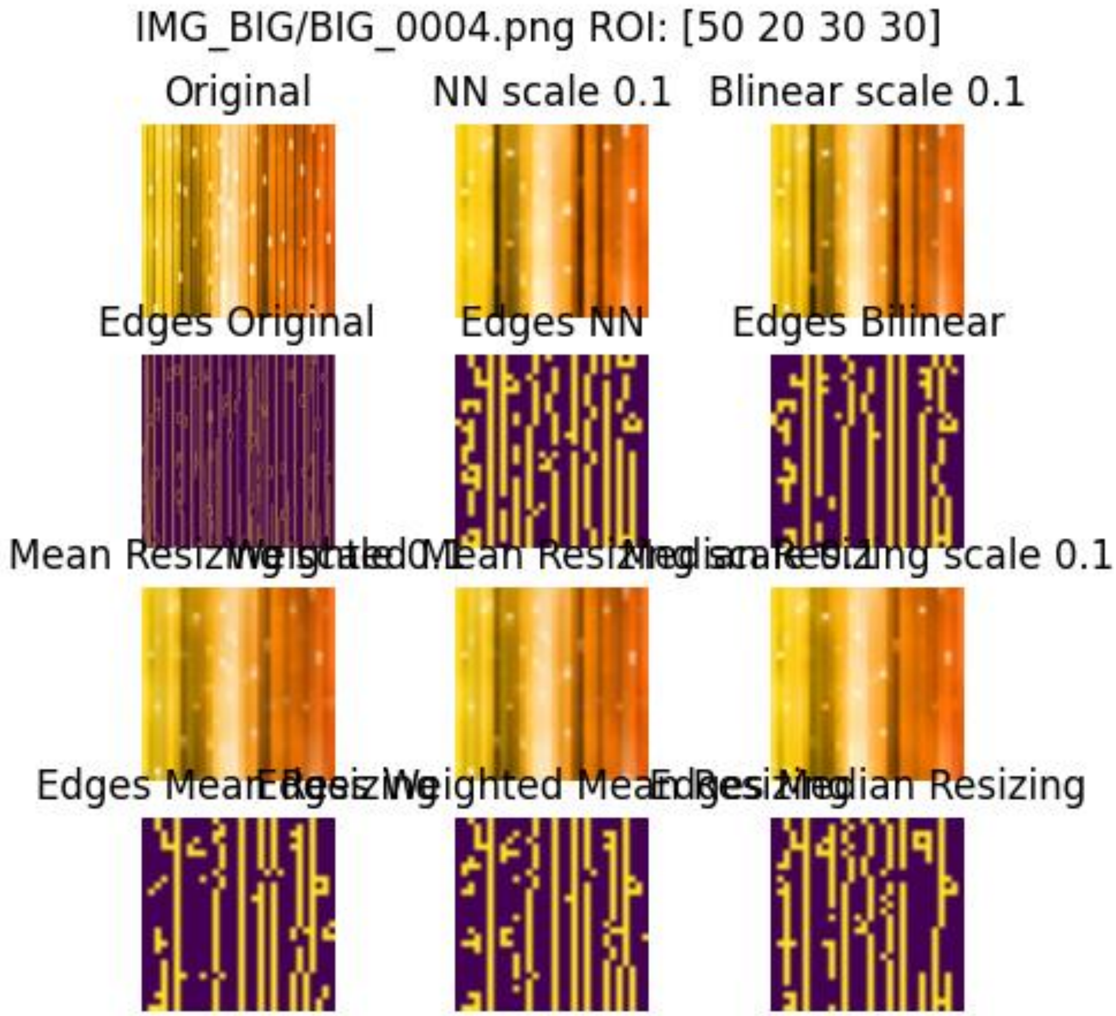
NN scale 0.1
Blinear scale 0.1
Edges Original
Edges NN
Edges Bilinear


IMG_BIG/BIG_0004.png ROI: [50 20 30 30]
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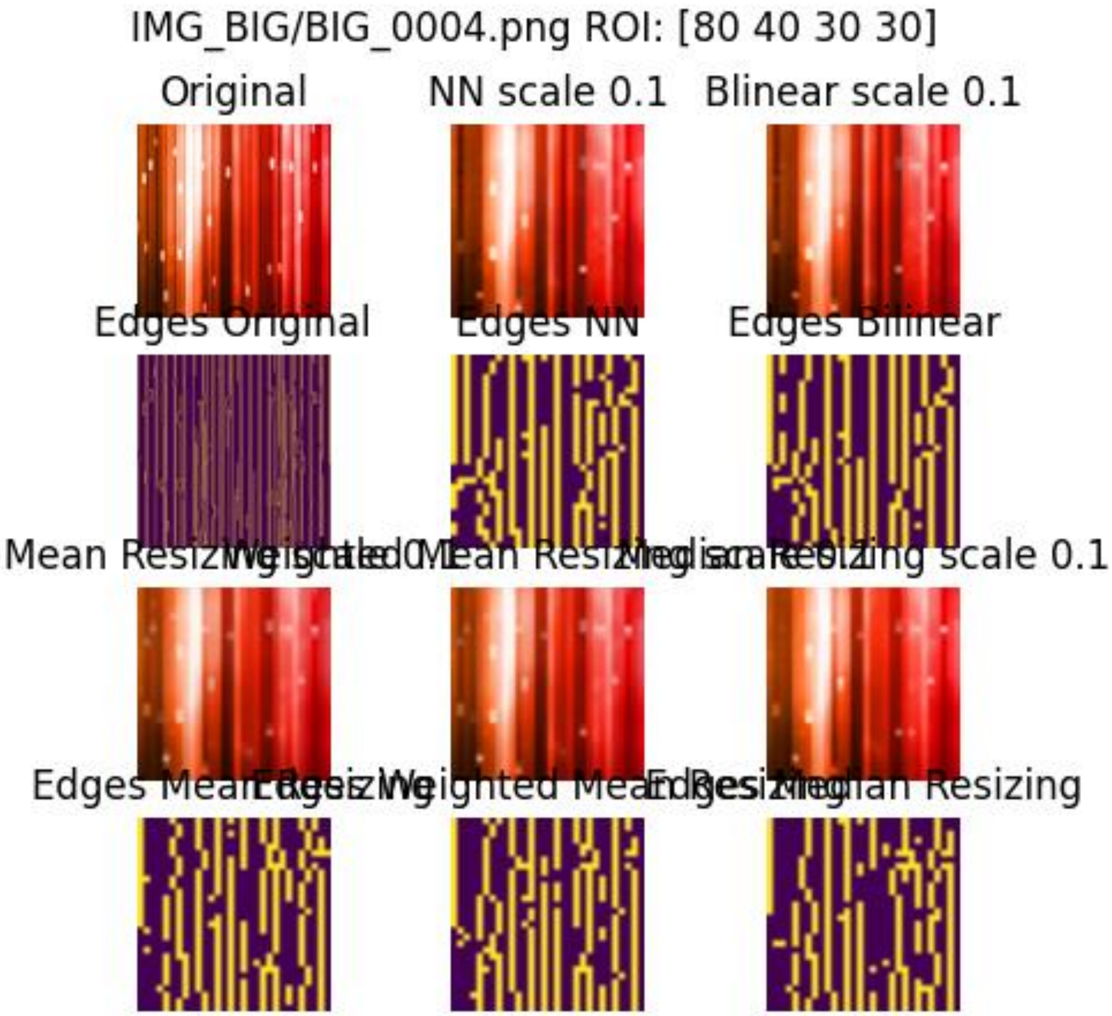
Blinear scale 0.1
Edges Original
Edges NN
Edges Bilinear


IMG_BIG/BIG_0004.png ROI: [80 40 30 30]
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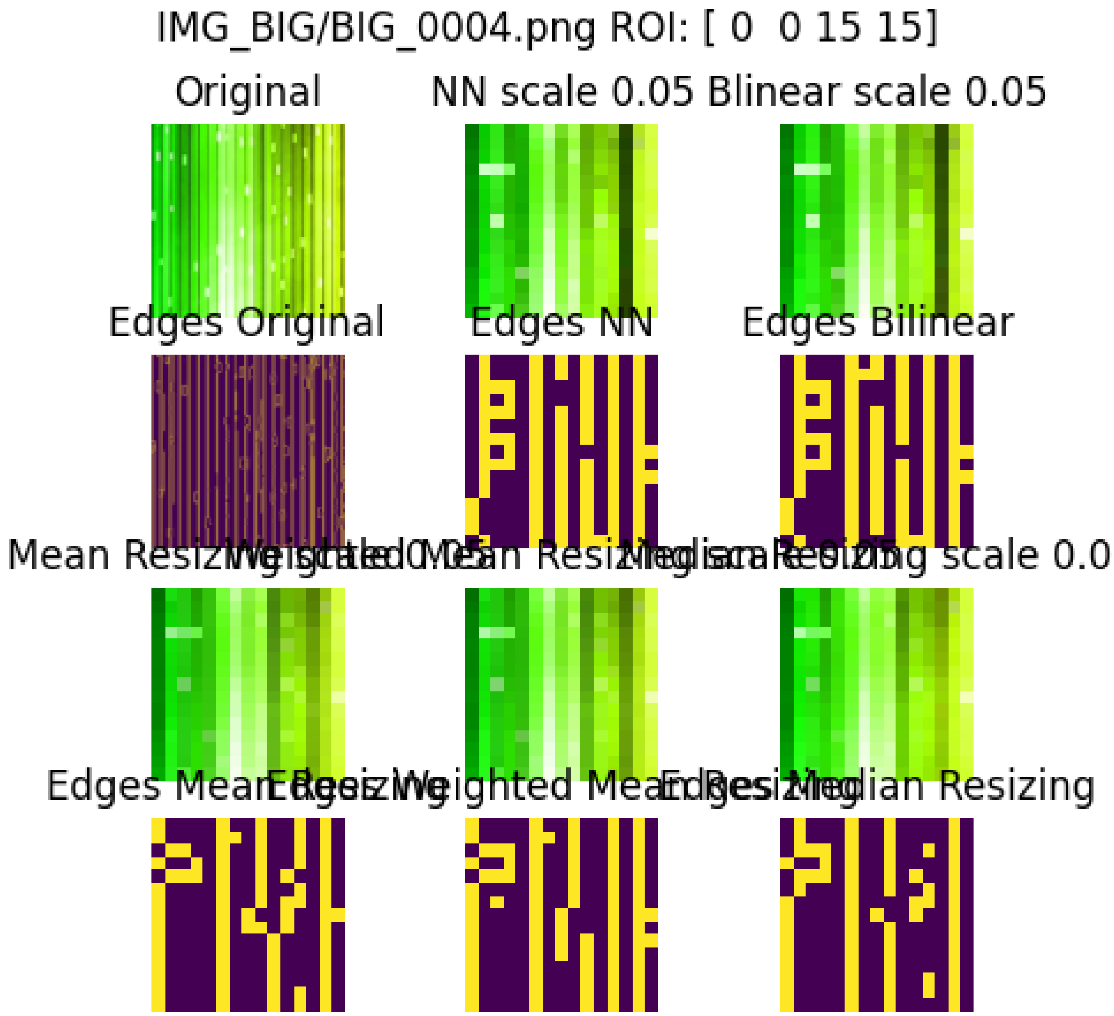
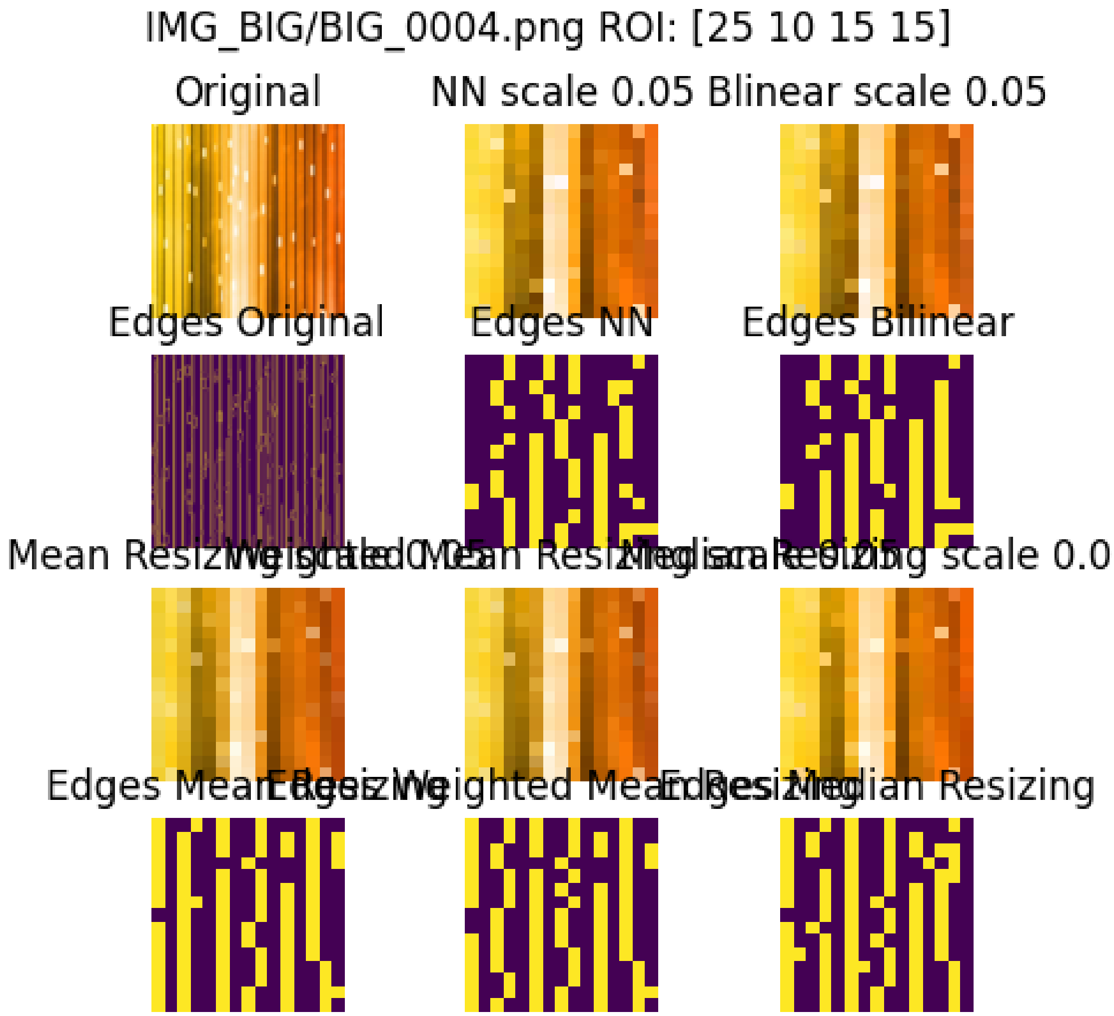
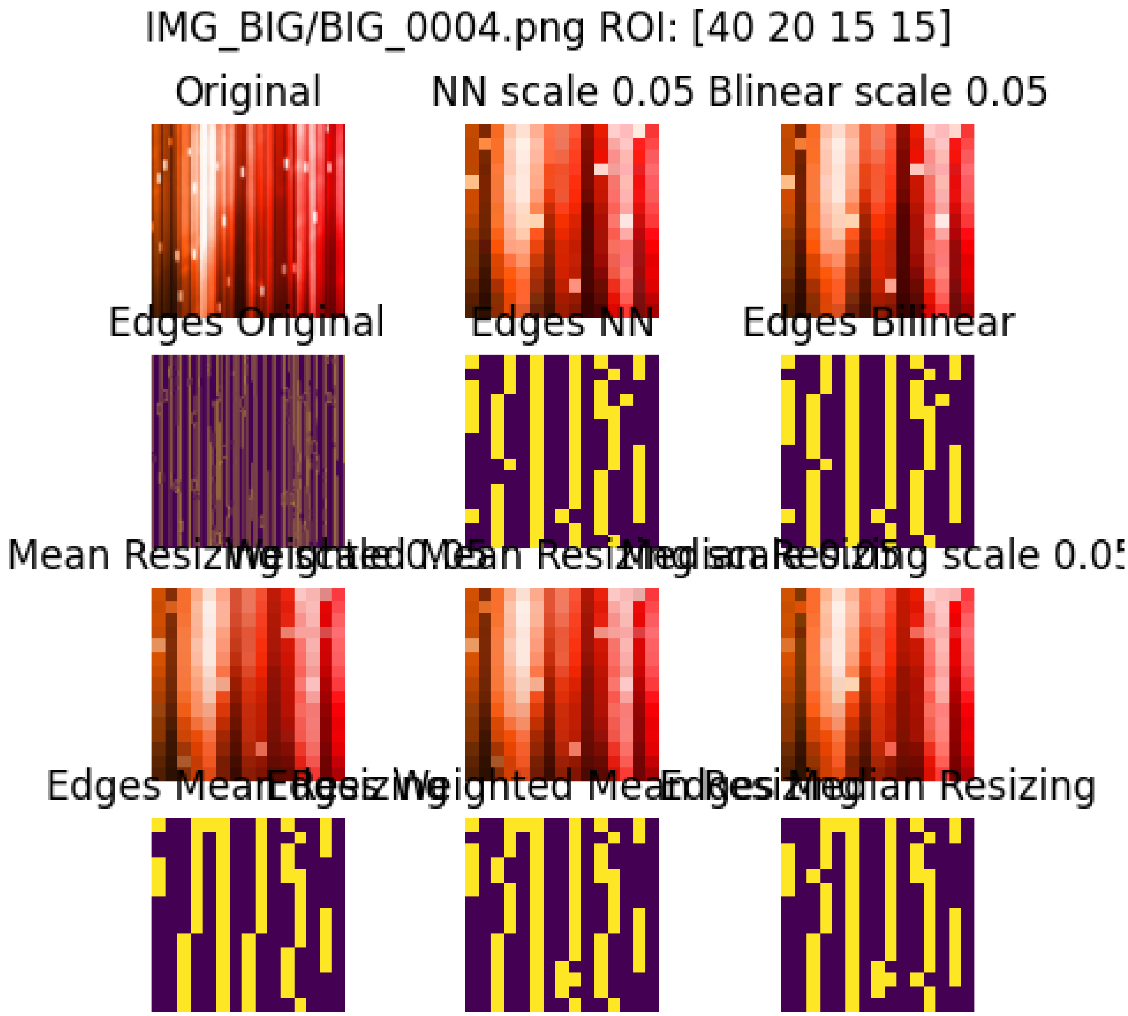
Blinear scale 0.1
Edges Original
Edges NN
Edges Bilinear


IMG_BIG/BIG_0004.png ROI: [0 0 15 15]
Original
NN scale 0.05
Blinear scale 0.05
Edges Original
Edges NN
Edges Bilinear
Mean Resizing scale 0.05
Weighted Mean Resizing scale 0.05
Median Resizing scale 0.05
Edges Mean Resizing
Edges Weighted Mean Resizing
Edges Median Resizing


IMG_BIG/BIG_0004.png ROI: [25 10 15 15]
Original
NN scale 0.05
Blinear scale 0.05
Edges Original
Edges NN
Edges Bilinear


IMG_BIG/BIG_0004.png ROI: [40 20 15 15]
Original
NN scale 0.05
Blinear scale 0.05
Edges Original
Edges NN
Edges Bilinear

Podsumowanie i wnioski

Tu proszę zebrać wszystkie obserwacje na podstawie powyższych wykresów i napisać wnioski.

Powiększanie: W przypadku powiększania metoda najbliższych sąsiadów działa lepiej od metody bilinearnej. Przy powiększeniu w skali 2 obydwie metody działają podobnie i, patrząc na detekcję krawędzi, nawet w przypadku zdjęcia „IMG_SMALL/SMALL_0001.tif" metoda bilinearna wykazała bardziej szczegółowy obraz, lecz w innych skalach powiększania (3, 5) metoda najbliższego sąsiada daje dużo bardziej widoczne i ostre krawędzie. Jednak jakość obrazków wciąż pozostaje na podobnym poziomie, porównywalnym do oryginału, ponieważ na wykresach różnice dokładnie widać dopiero po analizie krawędzi.

Pomniejszanie: W przypadku pomniejszania wszystkie metody prezentują podobną skuteczność, to znaczy detekcja krawędzi we wszystkich metodach daje podobne rezultaty, lecz w przypadku pomniejszania jakość zdjęć znacząco spada — zdjęcia stają się dużo mniej wyraźne.